

Technology and Information System (TIS) Licensing of TIS Software and Hardware

SUPPLIER PROPOSAL

Response to the IWF Request for Proposal dated 28 September 2026

Submission deadline: Saturday, 24 October 2026

Cover Letter

To: International Weightlifting Federation

Email: iwf@iwf.sport

Subject: IWF TIS RFP - Proposals

Dear Evaluation Committee,

[SUPPLIER NAME] submits this proposal in response to the International Weightlifting Federation's Request for Proposal dated 28 September 2026 for the licensing, supply and operation of complete Technology and Information Systems for weightlifting competitions.

[SUPPLIER NAME] has extensive experience providing sport technology services at international events, including World Championships and the Olympics, including the advanced integration profiles.

This proposal is based on the OTIS commercial offering, which includes the OWLCMS software, Blue Owl refereeing Devices, and the required computers to run a competition. This combination has managed extremely large competitions (5 simultaneous platforms with 1600 athletes) in full compliance with the IWF TCRR. The OWLCMS software has been in use since 2019 in its current form and has been used in over 107 countries. It is the de facto standard for state, national and continental competitions in the Pan-American confederation and regional federations like the Nordic federation. Over 60 kits of Blue Owl refereeing devices have been sold worldwide.

The proposal also specifies the additional computers and devices needed to meet the more advanced integration profiles.

We welcome the opportunity for certification of this offering.

Yours faithfully,

Authorisation	Details
Name	[TO BE COMPLETED]
Title	[TO BE COMPLETED]
Signature	[TO BE COMPLETED]
Date	[TO BE COMPLETED]

Proposal information	Supplier response
Supplier legal name	[TO BE COMPLETED]
Product / solution name	[TO BE COMPLETED]
Exact baseline	[TO BE COMPLETED]
Profiles claimed	[TO BE COMPLETED]
Primary contact	[TO BE COMPLETED]
Email / telephone	[TO BE COMPLETED]
Proposal date	[TO BE COMPLETED]
Document version	Draft 0.1

Revision Notes / TODO

Temporary working notes. Remove this section before final submission.

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A. Company

Legal name, registration, ownership/contact details, experience and references.

A.1 Supplier Eligibility and Responsibilities

A.1.1 Minimum supplier requirements

A.1.1.1 Legally established entity able to contract with IWF

[RFP §8.1] Legally established entity able to contract with IWF.

[To be completed]

A.1.1.2 Relevant experience delivering competition scoring/results or comparable mission-critical systems

[RFP §8.1] Relevant experience delivering competition scoring/results or comparable mission-critical systems.

[To be completed]

A.1.1.3 Ability to accept end-to-end responsibility for the complete system and declared third-party components.

[RFP §8.1] Ability to accept end-to-end responsibility for the complete system and declared third-party components.

[To be completed]

A.1.1.4 Qualified technical, competition-operations and support personnel

[RFP §8.1] Qualified technical, competition-operations and support personnel.

[To be completed]

A.1.1.5 International training, event support and incident-response capability

[RFP §8.1] International training, event support and incident-response capability.

[To be completed]

A.1.1.6 Ability to maintain the approved baseline for the license period

[RFP §8.1] Ability to maintain the approved baseline for the license period.

[To be completed]

B. Product

Product name, architecture, supported profiles, capacities, deployment model and limitations.

B.1 Solution Overview

This proposal relies on the OTIS kit described at <https://blue-owl.nemikor.com/otis>.

- The OWLCMS software, used worldwide at the Continental, Regional, National and Local levels. The software has been used in literally thousands of meets in 107 countries. OWLCMS is fully documented at <https://owlcms.jflamy.dev>.
- Blue Owl devices. Close to 70 kits of the devices proposed here have been sold. The devices are explicitly designed to work with OWLCMS. Some federations have deployed up to 6 platforms simultaneously with these devices.
- Additional computers and devices required for video processing and external interfaces are included to complement OTIS and meet the requirements of the advanced profiles.

Figure 1 and Figure 2 below provide a visual overview of the solution as proposed.

B.2 Profiles

B.2.1 M0 - Core Sport Conformance

[RFP §2.1] Complete sport logic, mandatory FOP controls, screens and official documents.

OWLCMS version 69 and Blue Owl version 3 are fully compliant with the TCRR as of October 2026.

B.2.2 M1 - Licensed Operational Baseline

[RFP §2.1] M0 plus baseline control, audit, interfaces, offline operation, recovery, procedures, training, support and spares.

The system as delivered will meet all these requirements. OWLCMS includes a module specifically meant for the creation of external data interfaces. The hardware devices use documented protocols.

B.2.3 M2 - Major Event & Broadcast

[RFP §2.1] M1 plus applicable VPT, public/live services, broadcast outputs, enhanced monitoring, high availability and full rehearsal.

Video playback uses video-capable equipment that is included in the proposal (laptops with video capture/gaming cards, cameras). OWLCMS has its own standalone VPT capabilities, and provides a core set of TV Graphics for broadcasts, and its replay streams are available to authorized consumers.

Furthermore, [Supplier Name] has extensive experience delivering broadcast quality TV Graphics and other interfaces.

B.2.4 M3 - Games & External Integration

[RFP §2.1] M2 plus approved Games/ORIS and programme-specific integrations.

Supplier has previously integrated OWLCMS with the ODF (Olympic Data Format) for Games integration in the past (Islamic Solidarity Games 2025). The OWLCMS interfacing protocol used for this purpose provides secure, read-only access to the competition data and competition updates, and can be used for any other similar integration need.

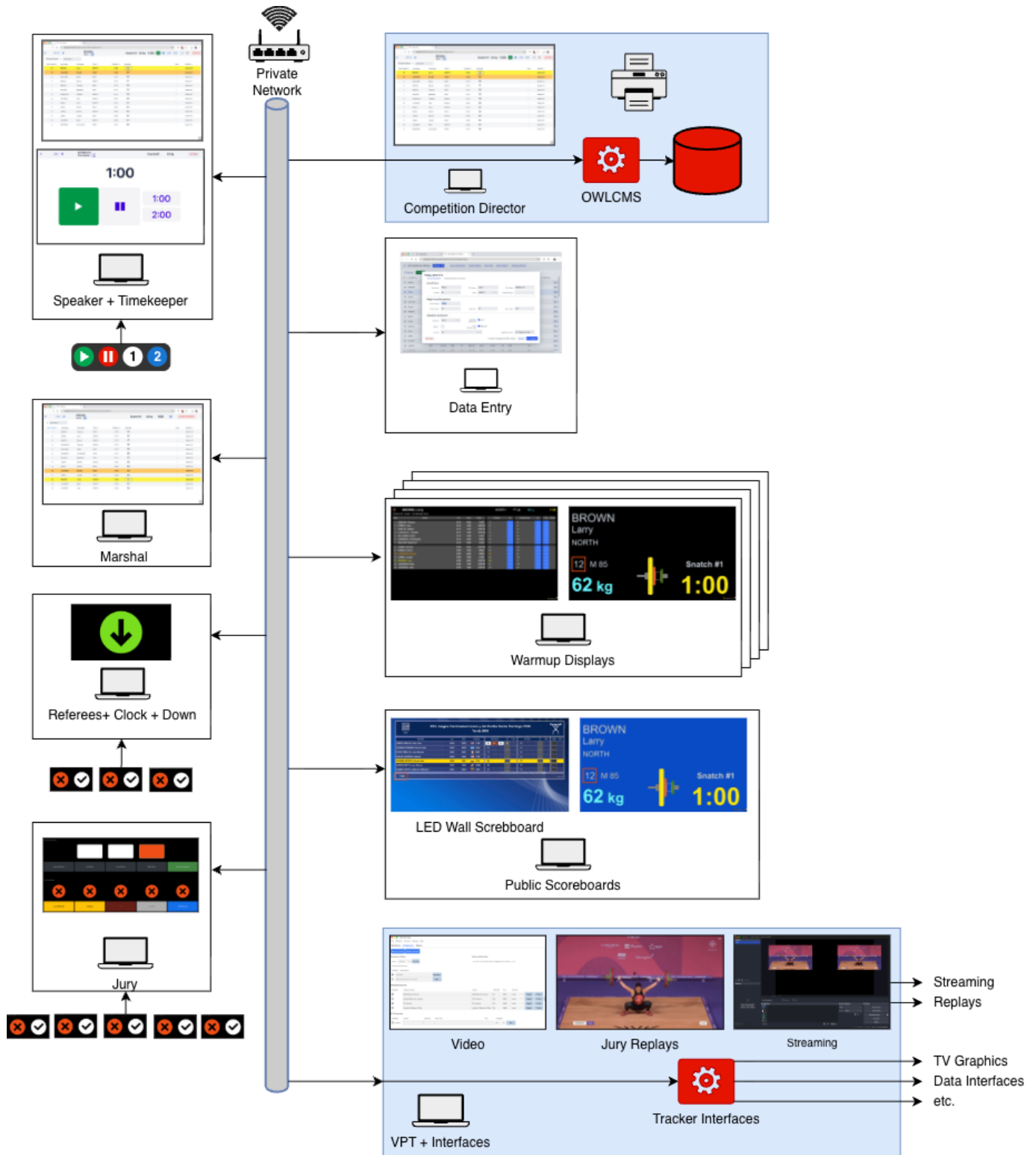


Figure 1: Solution Overview

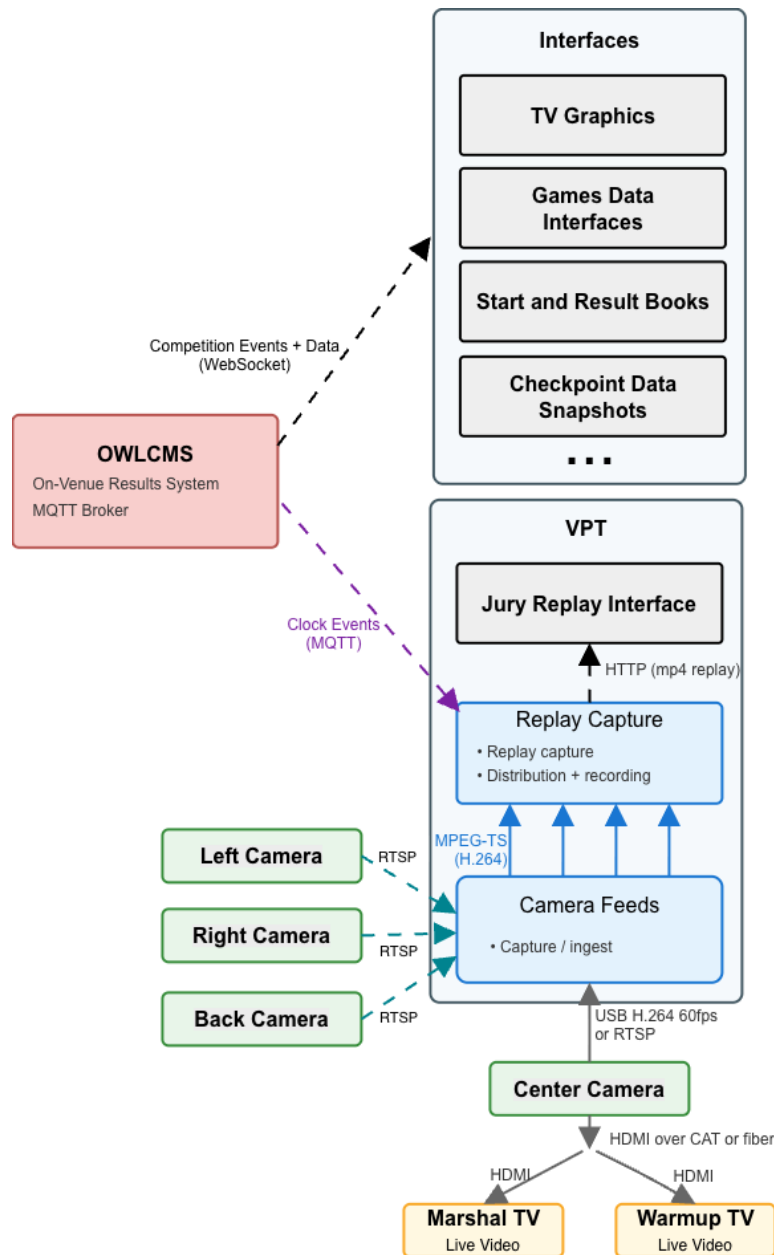


Figure 2: Detail: VPT and Interfaces

B.2.5 MP - Multi-platform Overlay

[RFP §2.1] Simultaneous platform operation with separation and capacity evidence.

OWLCMS has been used in actual competitions with up to 6 platforms simultaneously and 1600 athletes competing, with extremely low computer and network usage. The program is verified with a stress test that simulates competition updates as fast as possible in real life. Any competition can be simulated this way as soon as preliminary registration data is available.

When running multiple platforms, a single instance of OWLCMS runs them all together, and no integration step for results is required. All results and rankings are available live for all platforms. The normal constraints expressed in TCRR (group B should finish before group A for a given category) should nevertheless be respected so coaches get meaningful information.

B.3 Licence certificate

The certificate will state supplier, product and exact baseline, approved profiles, capacities, interfaces, critical dependencies, limitations, deviations, validity and surveillance conditions.

B.3.1 Supplier and product

[RFP §2.2] The licence certificate shall state the supplier and product.

The supplier of the product shall be [Supplier], who will act as integrator of the OTIS solution and of the necessary additional components required to meet profile M3.

B.3.2 Exact baseline

[RFP §2.2] The licence certificate shall state the exact baseline.

Any stable version of OWLCMS after version 69 is claimed to meet the current requirements and is fully compliant with TCRR as published at the time of this tender. Any version of the Blue Owl software after version 3.0 meets the requirements. Final baseline will be confirmed after acceptance testing.

B.3.3 Approved profiles and capacities

[RFP §2.2] The licence certificate shall state approved profiles and capacities.

M2 and MP profiles: The system complies with the M2 and MP profiles, up to 6 platforms. For the purpose of this license, the number of participants shall be 1800 or less. There is no intrinsic limitation on the number of participants; this number is given as a safe, previously observed figure with a margin of safety.

M3 profile: [Supplier] integrated the OVR into the Islamic Solidarity Games 2025 central platform using ODF (Olympic Data Format) protocols. [Supplier] can provide interfaces for the protocols used at given major events. Supplier has provided broadcast-quality graphics for 17 sports in extremely large events such as the 2026 Asian Games.

B.3.4 Interfaces and critical dependencies

[RFP §2.2] The licence certificate shall state interfaces and critical dependencies.

The system has no critical dependencies. All communications use the standard HTTP and MQTT protocols.

The specific MQTT messages used for the devices are fully documented. The physical interfaces for devices are USB and RJ-45 connectors. Only commercially available cabling (CAT-6, USB 2.0, HDMI) is used.

The protocol used to create additional interfaces for Data Exports, TV Graphics, etc. is public and has a published reference implementation.

B.3.5 Limitations and deviations

[RFP §2.2] The licence certificate shall state limitations and deviations.

None known.

B.3.6 Validity and surveillance conditions

[RFP §2.2] The licence certificate shall state validity and surveillance conditions.

Licensing authorizes the use of the OTIS equipment and OWLCMS software package; the equipment and software may not be used if the user does not hold such a license.

IWF is expected to acquire OTIS equipment and software licences from [Supplier] for IWF use at competitions that are directly under IWF authority. IWF shall not loan or sublease its own licenses. Other organizers of IWF events such as multi-sport games are required to acquire their own licenses, or enter into arrangements with [Supplier Name]

C. Software

Modules, versions, rule-pack approach, database, interfaces, deployment and update model.

C.1 Software, Data and Interfaces

C.1.1 Software

C.1.1.1 Implement the effective IWF rule pack and support controlled updates

[RFP §6.1] Implement the effective IWF rule pack and support controlled updates.

The solution is tested against the TCRR rules using automated tests. OWLCMS has over 350 automated tests covering rules, reports and program behaviour. Any new release is tested against regressions.

C.1.1.2 Competition-critical functions shall not depend on unavailable internet, licence-server or payment services.

[RFP §6.1] Competition-critical functions shall not depend on unavailable internet, licence-server or payment services.

Once downloaded, the solution is completely autonomous and works with no internet connection whatsoever

C.1.1.3 Exact production version, configuration, database schema, rule pack and relevant assets shall be identifiable and reproducible.

[RFP §6.1] Exact production version, configuration, database schema, rule pack and relevant assets shall be identifiable and reproducible.

Releases are created by a fully repeatable automated build process, and under strict version control. This applies both to OWLCMS and the Blue Owl components. All past versions and the source code used to build them are available.

C.1.1.4 Installation and rollback shall be documented and repeatable

[RFP §6.1] Installation and rollback shall be documented and repeatable.

OWLCMS is launched by a control panel that allows switching between versions. Within a major version, the database can be rolled back if a fix release is unsatisfactory. Fix releases can always read exported prior data so they can be tested independently. Emergency installation of a fix is a matter of seconds once the fix release is downloaded. The OWLCMS program runs on any available Windows, macOS or Linux computer, and the database exports are completely portable, so any available computer can be used for testing.

C.1.1.5 Automatic updates shall not occur during an event freeze unless approved

[RFP §6.1] Automatic updates shall not occur during an event freeze unless approved.

The system does not perform any type of automatic update. Operating system updates shall be managed by duly authorized personnel according to supplier procedures; they are not under the control of the solution.

C.1.2 Data integrity and authoritative state

C.1.2.1 Every sporting fact shall have one declared authoritative owner

[RFP §6.2] Every sporting fact shall have one declared authoritative owner.

Any change to the competition state is uniquely attributed to a human actor, to a device input, or to an automated calculation. The audit log reflects this attribution.

C.1.2.2 Derived screens, feeds, documents and caches shall be traceable to the accepted state

[RFP §6.2] Derived screens, feeds, documents and caches shall be traceable to the accepted state.

All computed results are directly attributable to programmed rules based on the authoritative data.

C.1.2.3 Corrections shall be authorised, atomic, attributable and propagated

[RFP §6.2] Corrections shall be authorised, atomic, attributable and propagated.

Once the competition has started, all changes are meant to be made using the interactive interfaces or the explicit import features of the programs such that audit traces are kept. Database state that does not correspond to audited

actions is evidence of tampering. Should intentional exceptional corrective actions by the supplier requiring database queries or similar be necessary, the corresponding authorizations are expected to be managed separately as part of incident response.

C.1.2.4 Duplicate, lost, conflicting or out-of-order transactions shall be prevented or detected.

[RFP §6.2] Duplicate, lost, conflicting or out-of-order transactions shall be prevented or detected.

The system applies strict sequencing for all events, whether device-generated or human-generated. Duplicate events are eliminated.

Conflicting updates from technical officials are detected and reported (e.g., competition director and marshal both attempting to correct an error). The marshals cannot accidentally interfere with the data updates resulting from a refereeing decision (the athlete card is closed when a decision is received). Athlete registration information cannot be edited while a session is in progress. Likewise, the features that enable post-session athlete card corrections are blocked until the session is over – during a session only the marshal and director pages can edit the athlete cards.

The screenshot shows a web application interface for a lifting competition. The main table lists athletes with columns for Start Number, Last Name, First Name, Team, Category, Attempts, Total, and Attempt. A modal dialog is open over the table, titled "Athlete changed since you opened this form". The dialog shows the current athlete being edited is "BROWN, Larry" and provides a comparison table between "When opened" and "Now" values. The dialog also includes buttons for "Save my values anyway", "Discard my changes", and "Keep editing".

Field	When opened	Now	Your form
snatch 1 – Change 1	–	65	63
snatch 1 – Change 2	–	67	–

C.1.2.5 A complete machine-readable event export shall be available to IWF

[RFP §6.2] A complete machine-readable event export shall be available to IWF.

The database export contains all the event data in structured JSON formats. The audit logs are also machine-readable.

C.2 Third-party dependencies

[RFP §6.4] All third-party software, libraries, services, licences and hardware necessary for operation shall be declared, including version, ownership, support status and cloud/public-internet dependency.

All third-party software, libraries, services, licences and hardware necessary for operation shall be declared, including version, ownership, support status and cloud/public-internet dependency.

All dependencies used in OWLCMS and Blue Owl are public and available on public servers.

C.2.1 OWLCMS dependencies

OWLCMS is written in Java. The full dependencies for OWLCMS are listed in the following files.

- <https://github.com/jflamy/owlcms4/blob/dev68/pom.xml>
- <https://github.com/jflamy/owlcms4/blob/dev68/owlcms/pom.xml>

All the dependency files are found in the industry-standard “Maven Central” repository for Java dependencies. Any past version of the software can be recreated because this site contains the full historical record of the libraries. The build process for OWLCMS is entirely automated and repeatable.

The main libraries in use are as follows

- The current long-term support version of the Vaadin user interface toolkit; currently version 25.2.7
- Extremely stable version of the Hibernate data access library, version 5.6.15.Final
- Extremely stable version of the H2 relational database, version 2.1.124
- Extremely stable version of the Eclipse Jetty Web Services, version 12.0.4
- Extremely stable version of the Eclipse Paho MQTT library (version 1.2.4)
- Extremely stable version of the Google Guava library for event distribution (version 33.5.0)
- The very common Apache POI library for generating Excel spreadsheets (Version 4.1.2)
- The jxls library for creating templates of Excel files for report generation using Apache POI (version 3.0.0)
- The very common logback library for writing journals and logging information
- The standard libraries for Java 25

Versions labelled as Extremely Stable have been in use for over 5 years without needing a change. We have applied all relevant security fixes to these libraries.

The libraries outside this list are utilities and do not represent architectural commitments (they can be substituted).

The only library subject to future updates is the Vaadin user interface library, to follow evolution in browser standards, usability improvements, and security fixes since browsers and associated technologies are often targeted.

C.2.2 Blue Owl Dependencies

Blue Owl is a Node.js application that connects the refereeing devices to OWLCMS. As such it depends on one main library to interact with the devices.

- The Firmata library johnny-five, version 2.1.0 (<https://github.com/rwaldron/johnny-five>)

C.2.3 Video Playback Dependencies

- The VPT module delegates all the actual video work to the very widely used FFmpeg software. On Windows the build used is the current stable version 9.0 (<https://github.com/BtbN/FFmpeg-Builds/releases/download/latest/ffmpeg-n9.0-latest-win64-gpl-shared-9.0.zip>)
- The specific version of FFmpeg used may change in the future depending on what graphics processor is present in the graphics processing laptop, and how device drivers evolve. Using FFmpeg isolates us from the compatibility details.

D. Hardware

Equipment list, specifications, quantities, spares and lifecycle/support information.

D.1 Hardware and Field-of-Play Requirements

D.1.1 Servers / control compute

[RFP §5] Controlled capacity for authoritative state and approved services, with spare/recovery path.

OWLCMS is extremely efficient in memory, input-output and networking. As a reference, the standard OTIS kit is tested to run 1600 athletes on 6 simultaneous platforms on a 8GB Raspberry Pi5 miniPC.

The proposed solution for IWF certification uses a standard 16GB Windows laptop with a SSD disk. That station is used by the competition director, and also includes Microsoft Office.

D.1.2 Operator stations

[RFP §5] Dedicated, role-controlled competition administration and operation positions.

Operator stations are only required to run two browsers as a maximum. The proposed configuration is that of the field-tested OTIS kit. Stations are 8GB Raspberry Pi500 that are cost-effective, easily replaceable, will be available as such for 15 years, and are designed to be used in difficult environments such as classrooms.

The USB ports on the operator stations are used to connect the Blue Owl refereeing devices, as shown in Figure 1 above.

Keyboards with numeric pads will be provided for the Marshal and Data entry stations.

D.1.3 Database / storage

[RFP §5] Protected authoritative storage, backup and tested recovery.

A fully relational database called H2 is used. H2 has been used without a single known failure since 2009, and is extremely quick. The database sizes of a full competition are very small (typically well below 2MB). Furthermore, OWLCMS preserves the ordering of all events, and the number of data entry operations is very low, so there is very little concurrency. The only significant queries are for results production, and then again, most of the data fits in main memory for caching.

The actual H2 binary file can be kept and is machine-independent, in addition to the structured JSON exports. The exports is the primary means of backup and snapshot. A readable consistent snapshot can be obtained at any time from the competition director station. The laptop that implements the interfaces can also pull a snapshot on demand, as a quick recovery capability.

Any reasonably recent laptop – in particular, the interface and graphics laptop -- can act as a cold spare (use the latest snapshot and bring it back in under a minute).

D.1.4 Referee controls

[RFP \$5] Three independent red/white referee control devices.

The Blue Owl referee controls include the two buttons, a visible LED, a sound, and a vibration for warning the referees. Referee controllers are cabled in order to avoid any concern with batteries.



D.1.5 Jury controls

[RFP \$5] Individual devices for applicable Jury members plus President controls.

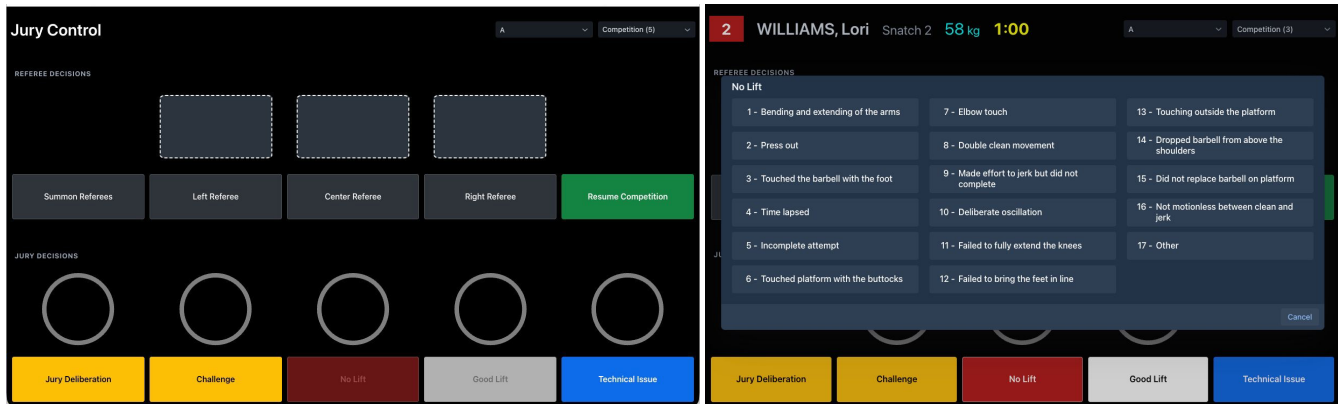
Jury controls for jury members are the same as the referee controls, but without the warning features. Referee controllers can be used as backups for jury controllers (the warnings would simply not be emitted).



President controls are of two kinds. The solution includes a physical keypad with all the required real-time referee decision display and the required jury member behaviour (green until all decisions visible)



The jury station will also have a tablet-style display that allows the jury president to indicate the reason for lift reversal.



D.1.6 Timing

[RFP \$5] Authoritative timer/controller with approved displays and audio.

Timing is done on the main server using the operating system timers. Operating system timing resolution is 1 ms as used. Propagation times of MQTT messages is typically under 10 ms.

Time is displayed on an athlete-facing display, on the attempt boards, and on scoreboards. Each machine with displays attached is synchronized with the OWLCMS computer and receives the millisecond timestamps at which OWLCMS emitted the command. In this way, the propagation delay (even though it is very small) can be compensated.



Timer warning and time-over signals are emitted on the down-signal device and logged in the audit log (for example, to determine if the timekeeper stopped the clock before or after the time-over buzzer).



D.1.7 Attempt boards

[RFP §5] Competition and warm-up presentation with mandatory athlete/attempt/weight/clock/decision data.

Attempt boards are standard FullHD or 4K TVs with HDMI inputs. The actual TV screens are expected to be provided locally and are not part of the kit.

As shown in Figure 1, two computers running standard web browsers provide the attempt boards as HDMI outputs. The Vaadin library sends the field of play updates to the browser over a WebSocket. There are two attempt board outputs

- The first attempt board output is meant to be customized by the supplier – typically the background color is adjusted to match the color scheme of the event. It is assumed that two main room attempt boards are present
- The second attempt board output is the normal high-contrast OWLCMS board and is used in all back-room areas (waiting, marshal, warmup)

The kit includes splitters and extenders to reach the typical waiting, marshal and warm-up areas located immediately behind the main platform. Further splitting or extension of the signal to reach multiple attempt boards will use HDMI or SDI equipment obtained locally.

Video connections for the main attempt boards would be as illustrated in Figure 3.

Figure 4 illustrates the back-room connections for attempt boards.

TODO: use pictures that reflect OTIS usage of dual-HDMI computers

TODO: use a more typical backroom layout

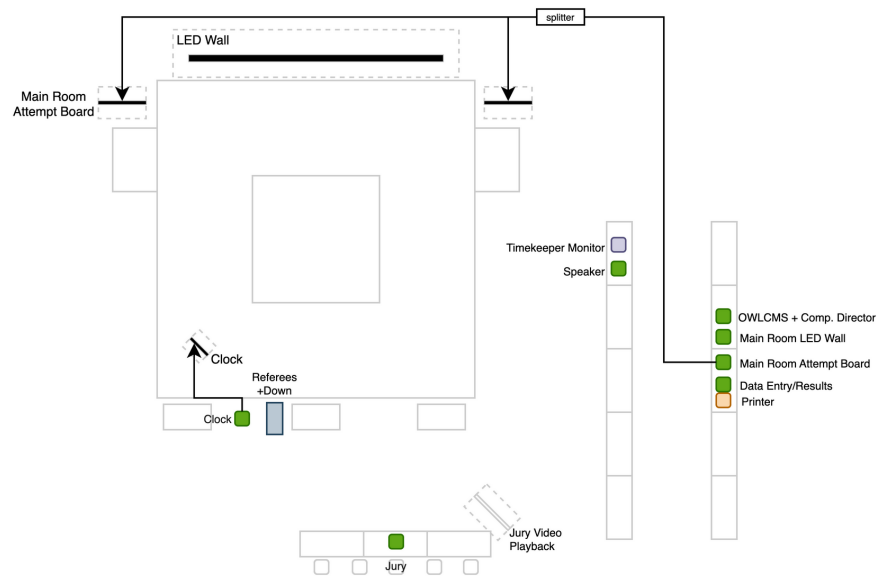


Figure 3: Attempt Board Video Connections

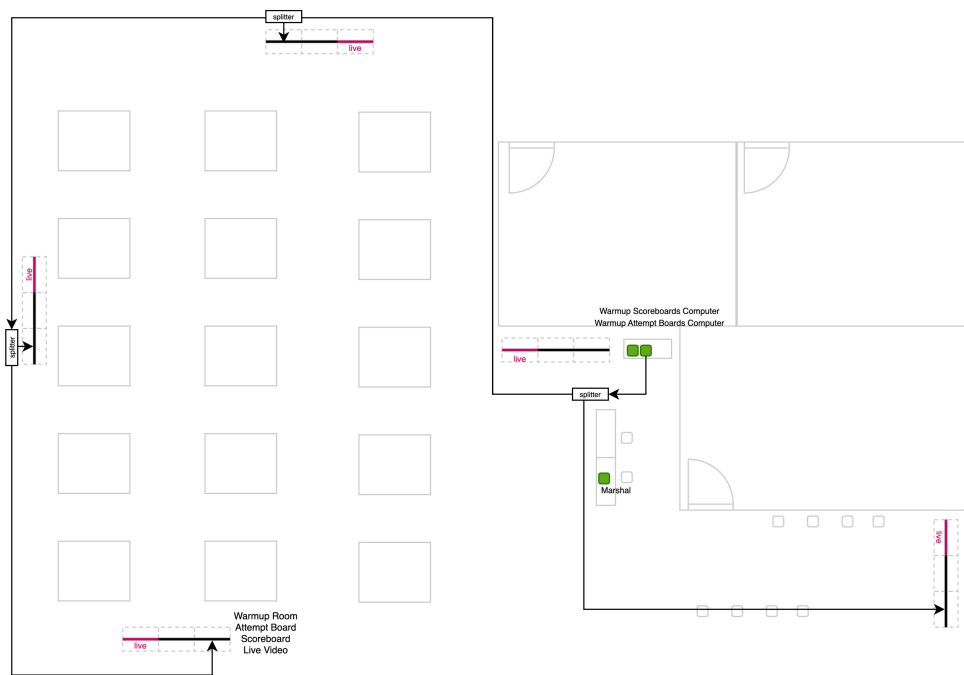


Figure 4: Warmup Room Attempt Board Video Connections

D.1.8 Scoreboards / RTDS

[RFP §5] Real-time competition and warm-up displays with correct ranking and state.

The situation for the scoreboards is the same as for attempt boards. There are two HDMI sources, one for the stage area, one for the back. For the stage area, the signal is assumed to go to a single large LED scoreboard.

D.1.9 VPT

[RFP §5] Approved cameras, recording/replay and Jury output for events requiring VPT.

The VPT system is under the full control of the competition staff, and the TCRR-mandated angles can be displayed in self-service mode by the jury if so required (or provided by the competition director).

As indicated in TCRR, the replay clips are accessible by the venue video team and broadcast teams. Such access will be controlled by networking equipment configurations.

The VPT system relies on four PTZ cameras. Based on our experience in Continental competitions, we propose Tenveo PTZ 60fps cameras. These cameras provide Power-over-Ethernet RTSP video feeds, HDMI, and compressed H.264 USB feeds.

OWLCMS has a VPT module that automatically clips and attributes the replays to the athlete lifting. This module keeps a full indexed archive of all the lifts, at all angles.

The VPT will execute on a high-performance laptop that also meets the requirements for the vMix software. If a client federation wants to add such additional video-processing capabilities, the proposed equipment will be sufficient for them to use their own license on the video-processing laptop.

D.1.10 Network

[RFP §5] Documented, segmented, monitored event network with common time.

The networking configuration is based on switching. Unmanaged switches are sufficient. A minimum of 4 POE (Power-over-Ethernet) ports is necessary to provide power to the networked PTZ cameras. A 16-port switch per platform is provided.

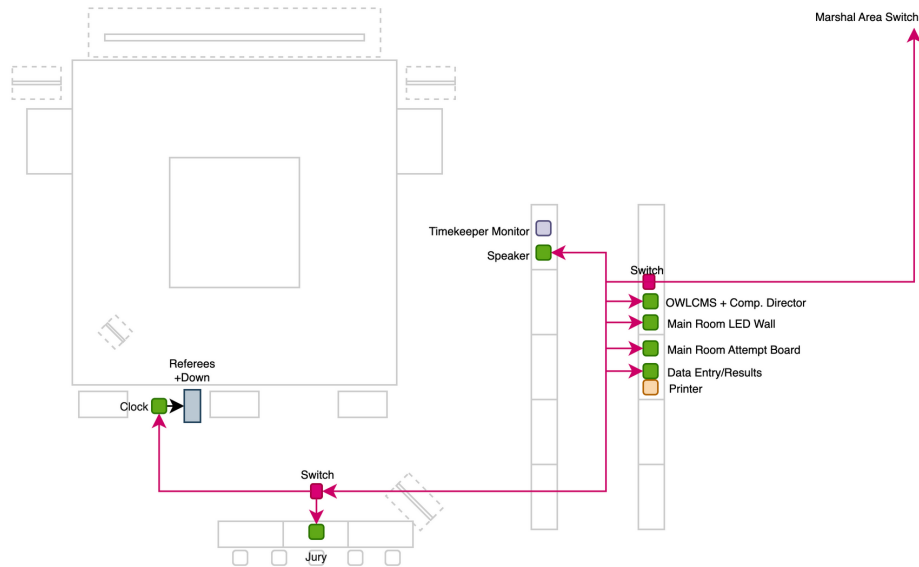


Figure 5: Main Room Networking

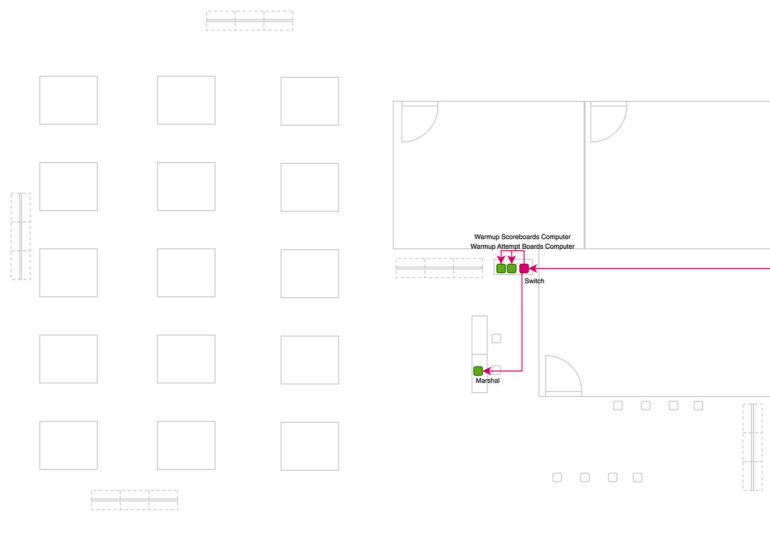


Figure 6: Back-room Networking

All the machines on one platform are connected to a switch on that platform. Use of a switch also ensures that video traffic from the cameras goes directly to the video equipment and does not touch other machines that don't need it

OWLCMS networking traffic is minimal. The application is routinely used in a cloud configuration with very remote servers and no perceptible delay.

An ASUS router is included. This router is used when an external wired internet connection is available for streaming. WiFi is not normally turned on. If federation personnel request access, it is preferred to loan them a network dongle. This eliminates concerns about illicit WiFi access.

D.1.11 Public / broadcast gateways

[RFP §5] Controlled read-only path from accepted state to approved consumers.

The video laptop will host all the external interfaces. In actual practice, sending information out to a games central is very low usage (and ODF protocols are push only). OWLCMS can also publish to cloud-based scoreboards and other feature, but those are also push only.

In the past, outside broadcasters have used the OVR-provided screen layouts directly. This was accommodated by opening a port on the router to a specifically white-listed address. This capability is generic with the equipment provided.

D.1.12 Printing

[RFP §5] Reliable official documents with fallback.

A Windows laptop with Microsoft Office is used for data entry and report generation to maximize compatibility with whatever printer is made available. OWLCMS produces documents in Excel format. Documents should be exported as PDF in non-editable mode and kept as such for archival.

Start books, Result books, Refereeing schedules and similar sophisticated layouts are produced as PDF.

D.1.13 UPS / protected power

[RFP §5] Protection and approved autonomy for critical equipment.

UPS will be provided to protect the networking switches in the front and back room, so the marshal and competition director can still capture changes and perform timing during a short power outage, relying on the fact that their stations have batteries.

Event organizing committees are expected to provide protection for the rest of the equipment, such as TVs and all other networking and video distribution, if warranted by local conditions.

D.1.14 Monitoring / logging

[RFP §5] Health, service, device, storage, time and state-state monitoring.

The OWLCMS software has very low memory and disk I/O requirements. Use of the normal operating system indicators is therefore sufficient. Additional technical logging for memory and CPU indicators can be turned on in the technical logs.

D.1.15 Spares / recovery media

[RFP §5] Rapidly deployable replacements for critical equipment families.

All the stations only require the presence of a current browser (Chrome, Edge, Firefox or Safari), and for some of them, the ability to connect a USB device. So essentially any laptop can be used, whether Windows, macOS, Linux. Chromebooks can be used when no USB device is needed.

OWLCMS runs on Windows, macOS,

D.1.16 FOP installation

[RFP §5] Safe, labelled cabling and equipment placement without obstruction or sightline conflicts.

[To be completed]

D.1.17 Hardware inventory

[RFP §5.1] For each licensed event, the supplier shall maintain make/model, quantity, serial or asset ID, firmware, location, connections, power source, owner, health status, spare path and approved deviations.

For each licensed event, the supplier shall maintain make/model, quantity, serial or asset ID, firmware, location, connections, power source, owner, health status, spare path and approved deviations.

Supplier response

[TO BE COMPLETED]

E. Competition functionality

Lifecycle, officiating, timing, displays, outputs, results and archive.

E.1 Minimum Functional Requirements –

The supplier claims full compliance with all the requirements. Details are provided for each item.

E.1.1 Data Intake

[RFP §3] Import and validate athlete, federation/NOC, official, event, venue and record data; stable IDs and error reporting.

Data is converted to an Excel-style sheet with the athlete identifier, name, and all registration data. Competition information and dates are stored in the program database (all information is stored in a relational database). The initial schedule is part of the data intake, and an initial assignment of athletes to sessions is done.

The input format handles multiple age groups and supports team allocation. The intake format can be exported, edited off-line, and re-ingested without loss of information. The same applies to schedule changes and referee assignments that can be modified in bulk until stabilized.

E.1.2 VFE / Technical Congress

[RFP §3] Corrections, approvals, random lots, grouping, timetable and Technical Official assignments.

The program prints VFE verification sheets that show exactly which athlete is in which category or categories. There are also team membership teams to verify that all team allocations are correct. These sheets can be signed by team representatives to facilitate the VFE process and reduce the number of errors, or used by the TIS team to do the capture and then get them signed.

A list of technical officials is kept, along with their team assignments. A referee team time table can be uploaded and then edited interactively. Individual assignment changes can then be made as required (e.g., substituting one referee for another in a given session).

E.1.3 Start publication

[RFP §3] Approved Start List and controlled distribution/versioning.

OWLCMS as proposed is able to produce a fully compliant start book. Preliminary versions can be published to list participants, with subsequent editions adding the schedule and the officials assignments. Once converted to the OWLCMS format, any number of records can be handled.

Field-of-play formats are also available (actual session assignments and schedule) for direct use by coaches and technical officials.

E.1.4 Weigh-in

[RFP §3] Correct timing, bodyweight, first attempts, 20 kg checks, start numbers, signatures and closed snapshot.

OWLCMS produces field-tested Weigh-in forms to capture both the weight-on-scale and deducted weight. All applicable rule are checked, including 20 kg and qualifying totals where applicable. The form captures all the required signature. Compliant athlete cards are likewise printed. A special data entry mode is used for quick capture of weigh-in data. A post-weigh-in snapshot of the database is taken.

Start numbers are cross-checked at the end of the weigh-in and checked against allocated bib numbers. A speaker introduction sheet for athletes and officials is produced, as are the empty protocols and jury member sheets.

E.1.5 Snatch

[RFP §3] Calling order, declarations/changes, 60/120-second timing, referee decisions, records and live outputs.

Full compliance with the competition flow is attained. All marshal data entry is immediately handled by the system and all lifting order or requested weights are instantly updated on all displays and scoreboards. All the normal timing decisions are automatic (the clock is automatically set to the right allocated time for each athlete, including on fast

sequences of changes). Only explicit loading incidents require a reset.

All attempts are checked against the 20 kg rule and the required changes are explicitly visible to the marshal.

Incomplete situations that do not fully restore the 20kg rule are explicitly visible.

E.1.6 Break

[RFP §3] Controlled Snatch-to-Clean & Jerk transition and synchronized displays.

Default rules for longer breaks (few athletes) and shorter breaks (large sessions) are both available. The session schedule also allows changing the break duration. This can also be done after the session has started (under the assumption the meet director informs all concerned).

Transition from the snatch to the clean & jerk break at the end of the snatch is automatic. A jury interruption or challenge on the last requires the time keeper starts the break once the situation has been resolved.

E.1.7 Clean & Jerk

[RFP §3] Calling order, weight changes, 20 kg enforcement, totals, ranks and records.

The same completely automatic timing and weight change flow as in the snatch takes place.

E.1.8 Review / correction

[RFP §3] STOP, Jury, challenge/VPT where applicable, additional attempts, correction, withdrawal and restart.

The jury can start a deliberation or accept a challenge using a physical jury keypad button or using a jury tablet. A flashing STOP sign is shown to the athlete on the platform, and all attempt boards and displays indicate the interruption. Transmission of the jury reversal code to the speaker can be done the traditional way (by the technical controller), or automatically when the jury president chooses the reversal code. In either case, the jury or challenge decision is shown as the speaker announces it.

The system can optionally display jury decisions on the scoreboards and the attempt boards. This capability is optional because it is not formally described in TCRR.

E.1.9 Results approval

[RFP §3] Protocol reconciliation, signatures/status, results, classifications and records.

A session protocol sheet is available immediately after the session that contains all the requisite information and each occurrence of an improved record. This is normally signed by the jury president. After a session, any correction required can be made via the system to ensure an audit trail to match whatever decisions were written on the protocols.

E.1.10 Ceremony / publication

[RFP §3] Medallists, public/broadcast outputs, communications and revised outputs.

The system distinguishes two types of scoreboards. Public scoreboards are normally styled to match the event colors, and switch content during ceremonies. OWLCMS will automatically switch during introductions. During medal ceremonies, the technical table (or the speaker themselves) can change the display to show the medals. In multi-age or multi-federation championships, this allows correct control of what is shown to the public. If broadcasters elect to use the video layout screens as provided by OWLCMS, the changes are also automatic.

E.1.11 Archive

[RFP §3] Final dataset, official outputs, logs, revisions, evidence and machine-readable export.

The system distinguishes two types of scoreboards. Public scoreboards are normally styled to match the event colors, and switch content during ceremonies. OWLCMS will automatically switch during introductions. During medal ceremonies, the technical table (or the speaker themselves) can change the display to show the medals. In multi-age or multi-federation championships, this allows correct control of what is shown to the public. If broadcasters elect to use the video layout screens as provided by OWLCMS, the changes are also automatic.

E.1.12 Core sporting logic

The core sporting logic is implemented is central logic called a “state automaton” that tracks all permissible actions. This logic in turn relies on computation rules (points, ranks, timing rules) that are automatically tested. There are over 300 automated tests that exercise the core sporting logic.

E.1.12.1 Snatch followed by Clean & Jerk, maximum three attempts per lift, correct Total calculation.

[RFP §3.1] Snatch followed by Clean & Jerk, maximum three attempts per lift, correct Total calculation.

Fully compliant with TCRR

Evidence / reference: Explicitly in database schema, automated tests.

E.1.12.2 Whole-kilogram weights and applicable minimum starting weights

[RFP §3.1] Whole-kilogram weights and applicable minimum starting weights.

Fully compliant with TCRR

Evidence / reference: Explicitly in database schema, automated tests.

E.1.12.3 Correct calling order

[RFP §3.1] Correct calling order, including weight, attempt number, previous-attempt sequence and start number.

Fully compliant with TCRR

Evidence / reference: Automated tests for all the change sequences.

E.1.12.4 Permitted weight changes, final-call and consecutive-attempt rules

[RFP §3.1] Permitted weight changes, final-call and consecutive-attempt rules.

Fully compliant with TCRR

Evidence / reference: Automated tests for all the change sequences and for the two-minute situations. Systematic recalculation of lift order to detect illegal moving down, with automatic tests.

E.1.12.5 Withdrawals, no-lift/no-Total conditions, disqualification and additional attempts

[RFP §3.1] Withdrawals, no-lift/no-Total conditions, disqualification and additional attempts.

Fully compliant with TCRR

Evidence / reference: Automated tests of the total, disqualification and withdrawals.

E.1.12.6 Material corrections recalculate affected rankings, records and outputs

[RFP §3.1] Material corrections recalculate affected rankings, records and outputs.

Fully compliant with TCRR

Evidence / reference: Any change to the lifting data causes a recalculation of all affected items. Automated tests for the ranking and records calculations.

E.2 Timing, Officiating and Outputs

E.2.1 Timing

E.2.1.1 Authoritative electronic countdown in one-second increments, configurable up to at least 15 minutes.

[RFP §4.1] Authoritative electronic countdown in one-second increments, configurable up to at least 15 minutes.

Fully compliant with TCRR

Evidence / reference: Timekeeper controls all break durations, no limit on duration.

E.2.1.2 Automatic 60- and 120-second attempt clocks and applicable introduction/break countdowns

[RFP §4.1] Automatic 60- and 120-second attempt clocks and applicable introduction/break countdowns.

Fully compliant with TCRR

Evidence / reference: Clock ownership is automatic, no intervention required to set the clock to 1:00, 2:00 or to the time remaining for the athlete. Transition between countdowns (introduction to snatch for example) is automatic. Can be overridden as required.

E.2.1.3 Correct start, stop, restart and STOP behaviour according to the rules

[RFP §4.1] Correct start, stop, restart and STOP behaviour according to the rules.

Fully compliant with TCRR

Evidence / reference: As per TCRR

E.2.1.4 All connected displays use the same authoritative time state within the approved tolerance.

[RFP §4.1] All connected displays use the same authoritative time state within the approved tolerance.

Fully compliant with TCRR

Evidence / reference: Time is kept on the OWLCMS server; when sent to remote devices, actual universal time reference for timer termination so the device can compensate for eventual small clock drift. All devices are time-synchronized.

E.2.1.5 Time corrections are authorised, recorded and auditable

[RFP §4.1] Time corrections are authorised, recorded and auditable.

Fully compliant with TCRR

Evidence / reference: The displays are NTP-synchronized at the operating system level by default. Verifying time synchronization is part of the technical setup.

E.2.2 Referee and Jury

E.2.2.1 Three independent referee control boxes with white/red decisions and required indicators.

[RFP §4.2] Three independent referee control boxes with white/red decisions and required indicators.

Fully compliant with TCRR

Evidence / reference: Nemikor referee buttons have visual indicator, sound, and vibration.

E.2.2.2 Correct majority/Down signal, reminder and reversal-window behaviour

[RFP §4.2] Correct majority/Down signal, reminder and reversal-window behaviour.

Fully compliant with TCRR

Evidence / reference: Implemented by software. Reversal rules are as per TCRR. Manual reversal of a decision following flags can easily be done by the technical table.

E.2.2.3 Jury visibility of individual referee decisions before public reveal

[RFP §4.2] Jury visibility of individual referee decisions before public reveal.

Fully compliant with TCRR

Evidence / reference: Live referee decisions and reversal are shown to the jury both on the keypad and on the jury console screen or tablet display if used.

E.2.2.4 Support for approved three- and five-member Jury configurations

[RFP §4.2] Support for approved three- and five-member Jury configurations.

Fully compliant with TCRR

Evidence / reference: Jury size is selectable,

E.2.2.5 Individual Jury decision controls and protected independence between members

[RFP §4.2] Individual Jury decision controls and protected independence between members.

Fully compliant with TCRR

Evidence / reference: Individual jury buttons are provided as per TCRR. Decision protection housings are not a TCRR requirement but will be addressed.

E.2.2.6 Material decisions and actions are attributable and time-stamped

[RFP §4.2] Material decisions and actions are attributable and time-stamped.

Fully compliant with TCRR

Evidence / reference: Full audit log of every single decision, timestamped to the millisecond, with the originating station or device.

E.2.3 Screens and live presentation

The system is fully compliant with all these items. Details and examples for each item are given.

NOTE: For many of the items, there are several screens available. We illustrate the key items and do not claim to be exhaustive.

E.2.3.1 System health

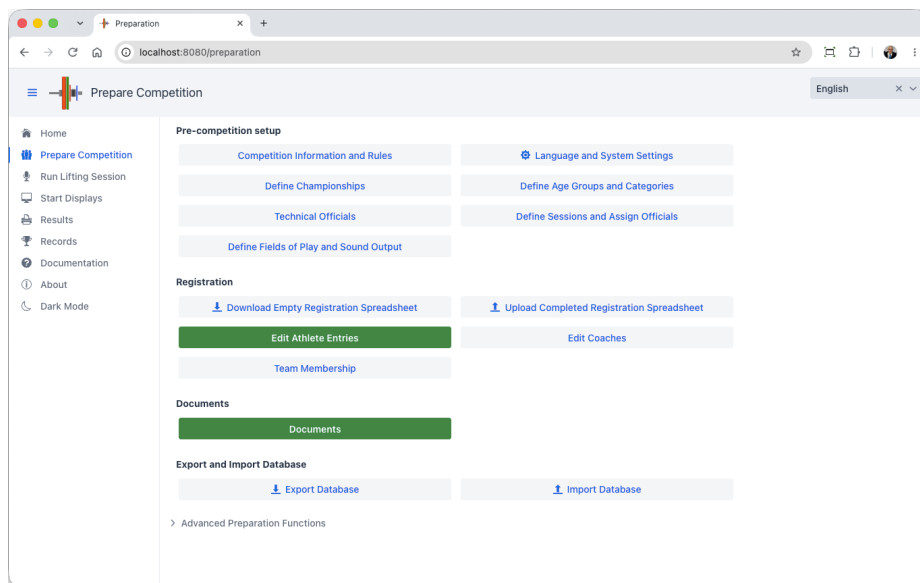
[RFP §4.3] At minimum, the TIS shall provide System health.

A technical log is always available for detailed technical operations. The memory, disk and networking usage are extremely low, the system gauges confirm this (the system never exceeds 20%, and is usually well below 10, even for extremely large meets).

E.2.3.2 Event setup

[RFP §4.3] At minimum, the TIS shall provide Event setup.

All competition information setup is done using interactive screens. Green is used in the software to indicate common actions.



E.2.3.2.1 Competition Rules

The first tab concerns competition information and which federation homologates it.

Edit Information and Rules

Information Competition Rules Default Scoring and Medaling Rules Scoreboard and Lifting Order

Competition

Name: Pan-American U15, U17 Championship, 1st PanAm Games Qualifier

Date: 2026-08-26 to 2026-08-30

Site: Guayaquil

Organizer: Federación Ecuatoriana de Levantamiento de Pesas

City: Ecuador

Federation

Federation Name: Federación Panamericana de Levantamiento de Pesas

Federation Address:

Federation e-mail:

Federation Web Site:

[Update](#)

Other Tabs contain the default rules for the meet, for example, what medals are given, how best athletes are determined, how team points are given, etc.

Edit Information and Rules

Information Competition Rules **Default Scoring and Medaling Rules** Scoreboard and Lifting Order

These default settings can be overridden for individual championships [Edit Championships](#)

Medals

Medals Awarded: ☒ Lifts and Total ☐ Total only ☐ Lifts only

Total Medal Scoring: Total

Scoring Systems

Best Athlete Scoring System: GAMX

Best CJ Scoring System:

Team Points

Placings counted: ☒ Lifts and Total ☐ Total only ☐ Lifts only

Team Points for 1st place: 28

Team Points for 2nd place: 25

Team Points for 3rd place: 23

Men Women Teams

Team Ranking Method: ☒ Sum of Points ☐ Sum of Scores

Team Scoring System: GAMX

Athletes per: 8

Maximum: 2

[Update](#)

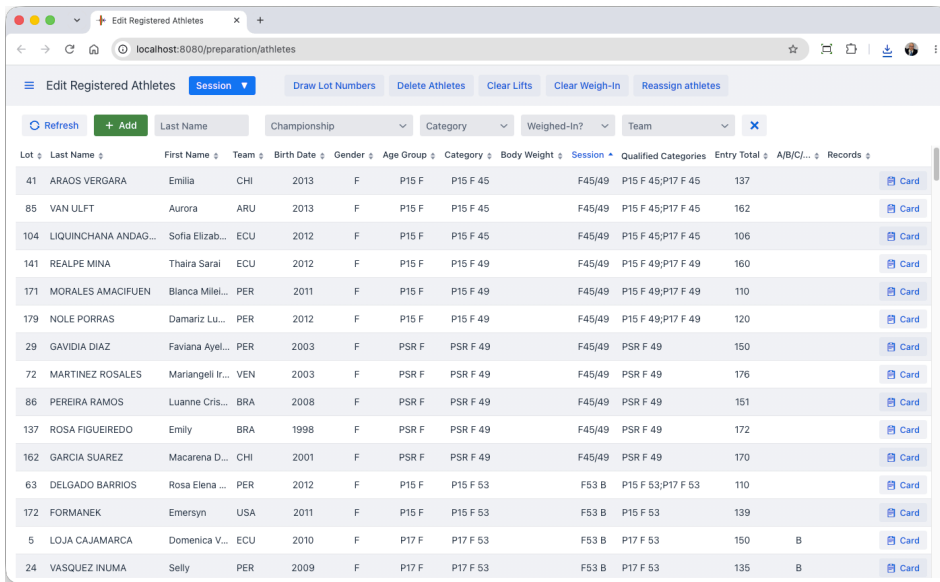
OWLCMS can also be used for local, national, regional and continental championships that have historical variations with full IWF rules. The other tabs include all these options. The defaults comply with TCRR but acquiring organizations would nevertheless be able to use their licenses in different settings.

E.2.3.2.2 Lots/groups/timetable

[RFP §4.3] At minimum, the TIS shall provide lots/groups/timetable.

Normally an import from a competitions registration system is used to set up the initial schedule and assign athletes assigned to A/B/C... groups. At the scale of IWF events, this is most likely done using output of the registration system in spreadsheet format. This is then converted to fit the documented OWLCMS format (this has now become trivial with human assistance tools). All data exchange formats for OWLCMS are public.

Once this is done all athletes can be edited in the program, and late entries added by hand if needed.

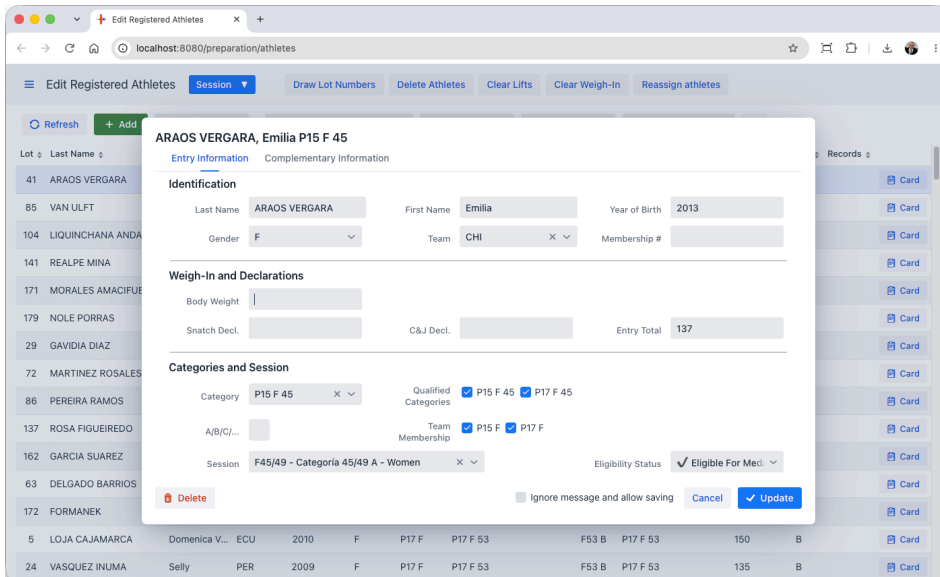


The screenshot shows a web application titled 'Edit Registered Athletes'. It features a navigation bar with tabs: 'Session', 'Draw Lot Numbers', 'Delete Athletes', 'Clear Lifts', 'Clear Weigh-In', and 'Reassign athletes'. Below the navigation bar is a search and filter section with fields for 'Last Name', 'Championship', 'Category', 'Weighed-in?', and 'Team'. The main content is a table of athletes with columns: Lot, Last Name, First Name, Team, Birth Date, Gender, Age Group, Category, Body Weight, Session, Qualified Categories, Entry Total, A/B/C/J..., and Records. The table lists 24 athletes, including ARAOS VERGARA, VAN ULFT, LIQUINCHANA ANDAG..., REALPE MINA, MORALES AMACIFUEN, NOLE PORRAS, GAVIDIA DIAZ, MARTINEZ ROSALES, PEREIRA RAMOS, ROSA FIGUEIREDO, GARCIA SUAREZ, DELGADO BARRIOS, FORMANEK, LOJA CAJAMARCA, and VASQUEZ INUMA. Each athlete's row includes a 'Card' link.

Lot	Last Name	First Name	Team	Birth Date	Gender	Age Group	Category	Body Weight	Session	Qualified Categories	Entry Total	A/B/C/J...	Records
41	ARAOS VERGARA	Emilia	CHI	2013	F	P15 F	P15 F 45	F45/49	P15 F 45;P17 F 45	137			Card
85	VAN ULFT	Aurora	ARU	2013	F	P15 F	P15 F 45	F45/49	P15 F 45;P17 F 45	162			Card
104	LIQUINCHANA ANDAG...	Sofia Elizab...	ECU	2012	F	P15 F	P15 F 45	F45/49	P15 F 45;P17 F 45	106			Card
141	REALPE MINA	Thaira Sarai	ECU	2012	F	P15 F	P15 F 49	F45/49	P15 F 49;P17 F 49	160			Card
171	MORALES AMACIFUEN	Blanca Miel...	PER	2011	F	P15 F	P15 F 49	F45/49	P15 F 49;P17 F 49	110			Card
179	NOLE PORRAS	Damariz Lu...	PER	2012	F	P15 F	P15 F 49	F45/49	P15 F 49;P17 F 49	120			Card
29	GAVIDIA DIAZ	Faviana Ayel...	PER	2003	F	PSR F	PSR F 49	F45/49	PSR F 49	150			Card
72	MARTINEZ ROSALES	Mariangeli Ir...	VEN	2003	F	PSR F	PSR F 49	F45/49	PSR F 49	176			Card
86	PEREIRA RAMOS	Luanne Cris...	BRA	2008	F	PSR F	PSR F 49	F45/49	PSR F 49	151			Card
137	ROSA FIGUEIREDO	Emily	BRA	1998	F	PSR F	PSR F 49	F45/49	PSR F 49	172			Card
162	GARCIA SUAREZ	Macarena D...	CHI	2001	F	PSR F	PSR F 49	F45/49	PSR F 49	170			Card
63	DELGADO BARRIOS	Rosa Elena ...	PER	2012	F	P15 F	P15 F 53	F53 B	P15 F 53;P17 F 53	110			Card
172	FORMANEK	Emersyn	USA	2011	F	P15 F	P15 F 53	F53 B	P15 F 53	139			Card
5	LOJA CAJAMARCA	Domenica V...	ECU	2010	F	P17 F	P17 F 53	F53 B	P17 F 53	150	B		Card
24	VASQUEZ INUMA	Selly	PER	2009	F	P17 F	P17 F 53	F53 B	P17 F 53	135	B		Card

The registration screen allows modification of the allocated session, moving the athlete to another group, adjusting eligibility to multiple age groups as appropriate.

The other tabs on the registration form cover things like record eligibility (in joint jurisdiction championships, not all athletes are eligible to improve all records)



The screenshot shows the 'Edit Athlete' modal for ARAOS VERGARA, Emilia. The modal is titled 'ARAOS VERGARA, Emilia P15 F 45' and has two tabs: 'Entry Information' and 'Complementary Information'. The 'Entry Information' tab is active and contains the following fields:

- Identification:** Last Name (ARAOS VERGARA), First Name (Emilia), Year of Birth (2013), Gender (F), Team (CHI), Membership #.
- Weigh-in and Declarations:** Body Weight, Snatch Decl., C&J Decl., Entry Total (137).
- Categories and Session:** Category (P15 F 45), Qualified Categories (P15 F 45, P17 F 45), Team Membership (P15 F, P17 F), Session (F45/49 - Categoría 45/49 A - Women), Eligibility Status (Eligible For Med).

At the bottom of the modal are buttons for 'Delete', 'Ignore message and allow saving', 'Cancel', and 'Update'.

The session timetable is editable.

Name	Description	Size	Time before Clea...	Weigh-in Time	Start Time	Platform		
F45/49	Categoría 45/49 A - Women	11	10	2026-08-26 09:00	2026-08-26 11:00	B	Edit Details	Edit Athlete Entries
F53 B	Categoría 53 B - Women	9	10	2026-08-26 11:30	2026-08-26 13:30	B	Edit Details	Edit Athlete Entries
M55/60	Categoría 55/60 A - Men	9	10	2026-08-26 15:00	2026-08-26 17:00	B	Edit Details	Edit Athlete Entries
F53 A	Categoría 53 A - Femenino	8	10	2026-08-27 09:00	2026-08-27 11:00	A	Edit Details	Edit Athlete Entries
M65	Categoría 65 A - Men	14	5	2026-08-27 11:00	2026-08-27 13:00	A	Edit Details	Edit Athlete Entries
F57	Categoría 57 A - Women	8	10	2026-08-27 13:30	2026-08-27 15:30	A	Edit Details	Edit Athlete Entries
M70	Categoría 70 A - Men	6	10	2026-08-27 15:30	2026-08-27 17:30	B	Edit Details	Edit Athlete Entries
F61 B	Categoría 61 B - Women	8	10	2026-08-27 17:00	2026-08-27 19:00	A	Edit Details	Edit Athlete Entries
F69 B	Categoría 69 B - Women	9	10	2026-08-28 10:00	2026-08-28 12:00	A	Edit Details	Edit Athlete Entries
F61 A	Categoría 61A - Women	8	10	2026-08-28 12:00	2026-08-28 14:00	B	Edit Details	Edit Athlete Entries
M75	Categoría 75 A - Men	13	10	2026-08-28 14:00	2026-08-28 16:00	A	Edit Details	Edit Athlete Entries
F69 A	Categoría 69 A - Women	9	10	2026-08-28 16:30	2026-08-28 18:30	B	Edit Details	Edit Athlete Entries
M85 A	Categoría 85 A - Men	13	10	2026-08-29 08:00	2026-08-29 10:00	B	Edit Details	Edit Athlete Entries
F77	Categoría 77 A - Women	11	10	2026-08-29 10:30	2026-08-29 12:30	B	Edit Details	Edit Athlete Entries

E.2.3.2.3 Assignments of Technical Officials

Add Timetable screen here

Adjustments to individual TO assignments can be done from here after the initial TO teams have been assigned.

Session F45/49

Session Officials Support Jury

Announcer

Colon Arrollo, David X

Timekeeper

Flores Fernandez, German Alejandro X

Marshal

Peralta Latorre, Natalia X

Marshal 2

Soto Osco, Margaret Duguid X

Technical Controller

Granados Benavides, Jorge Alberto X

Technical Controller 2

Ramos Vera, Juan X

Referee 1

Zavala Plaza, Melvin Jose X

Referee 2

Hardie, Marissa X

Referee 3

Solis Gonzalez, Julio Ulises Oliver X

Reserve Referee

X

Delete Cancel Update

E.2.3.3 Document Production

After registration and VFE, the relevant documents can be printed, such as the athlete cards and weigh-in forms. Documents are produced in Excel, and can therefore be easily converted to PDF if needed. All the necessary documents are available (see the Documents section further down)

Groups of documents used together can also be created as one set (for example, the weigh-in sheet and the athlete cards), for one or more session (for example, for the next day).

Name	Description	Size	Weigh-in Time	Start Time	Platform
F45/49	Categoria 45/49 A - Women	11	2026-08-26 09:00	2026-08-26 11:00	B
F53 B	Categoria 53 B - Women	9	2026-08-26 11:30	2026-08-26 13:30	B
M55/60	Categoria 55/60 A - Men	9	2026-08-26 15:00	2026-08-26 17:00	B
F53 A	Categoria 53 A - Femenino	8	2026-08-27 09:00	2026-08-27 11:00	A
M65	Categoria 65 A - Men	14	2026-08-27 11:00	2026-08-27 13:00	A
F57	Categoria 57 A - Women	8	2026-08-27 13:30	2026-08-27 15:30	A
M70	Categoria 70 A - Men	6	2026-08-27 15:30	2026-08-27 17:30	B
F61 B	Categoria 61 B - Women	8	2026-08-27 17:00	2026-08-27 19:00	A
F69 B	Categoria 69 B - Women	9	2026-08-28 10:00	2026-08-28 12:00	A
F61 A	Categoria 61 A - Women	8	2026-08-28 12:00	2026-08-28 14:00	B
M75	Categoria 75 A - Men	13	2026-08-28 14:00	2026-08-28 16:00	A
F69 A	Categoria 69 A - Women	9	2026-08-28 16:30	2026-08-28 18:30	B
M85 A	Categoria 85 A - Men	13	2026-08-29 08:00	2026-08-29 10:00	B
F77	Categoria 77 A - Women	11	2026-08-29 10:30	2026-08-29 12:30	B

E.2.3.4 Weigh-in

[RFP §4.3] At minimum, the TIS shall provide Weigh-in.

The weigh-in data entry form is intentionally the same format as the registration form. All relevant checks for the 20 kg rule are applied at data entry, in case they were missed by the weigh-in Tos.

ARAOS VERGARA, Emilia P15 F 45

Identification

Last Name: ARAOS VERGARA, First Name: Emilia, Year of Birth: 2013, Gender: F, Team: CHI, Membership #:

Weigh-In and Declarations

Body Weight: 44.9

Snatch Decl.: 50, C&J Decl.: 65, Entry Total: 137

Starting weight rule violation: This must be corrected before the athlete lifts. Missing 7 kg.

Categories and Session

Category: P15 F 45, Qualified Categories: P15 F 45, P17 F 45, Team Membership: P15 F, P17 F, Session: F45/49 - Categoria 45/49 A - Women, Eligibility Status: Eligible For Med.

Buttons: Delete, Ignore message and allow saving, Cancel, Update

After weigh-in, the page allows the creation of all the relevant forms for the speaker and for the jury.

Start#	Last Name	First Name	Team	Gender	Age Group	Session	Category	Body Weight	Snatch	C&J Decl.	Qualified Categories	Entry Total	Record Eligibility
0	ARAOS VE...	Emilia	CHI	F	P15 F	F45/49	P15 F 45				[P15 F 45, P17 F 45]	137	
0	VAN ULFT	Aurora	ARU	F	P15 F	F45/49	P15 F 45				[P15 F 45, P17 F 45]	162	
0	LIQUINCHANA...	Sofia Eliz...	ECU	F	P15 F	F45/49	P15 F 45				[P15 F 45, P17 F 45]	106	
0	REALPE M...	Thaira Saral	ECU	F	P15 F	F45/49	P15 F 49				[P15 F 49, P17 F 49]	160	
0	MORALES...	Blanca Mil...	PER	F	P15 F	F45/49	P15 F 49				[P15 F 49, P17 F 49]	110	
0	NOLE PO...	Damariz L...	PER	F	P15 F	F45/49	P15 F 49				[P15 F 49, P17 F 49]	120	
0	GAVIDIA D...	Faviana Ay...	PER	F	PSR F	F45/49	PSR F 49				[PSR F 49]	150	
0	MARTINE...	Mariangeli...	VEN	F	PSR F	F45/49	PSR F 49				[PSR F 49]	176	
0	PEREIRA R...	Luanne Cr...	BRA	F	PSR F	F45/49	PSR F 49				[PSR F 49]	151	
0	ROSA FIG...	Emily	BRA	F	PSR F	F45/49	PSR F 49				[PSR F 49]	172	
0	GARCIA S...	Macarena ...	CHI	F	PSR F	F45/49	PSR F 49				[PSR F 49]	170	

E.2.3.5 Chief Marshal

[RFP §4.3] At minimum, the TIS shall provide Chief Marshal.

The marshal table has a station that shows all athletes (in start order or lifting order as preferred by the TO operating). Clicking on an athlete brings up a card that is laid out exactly as the athlete card, and automatically brings focus to the right cell, such that typing and hitting Enter can be used for fast data entry.

Start Number	Last Name	First Name	Team	Category	Attempts	Total	Attempt
1	ARAOS VERGARA	Emilia				-	Snatch #1
2	VAN ULFT	Aurora				-	Snatch #1
3	LIQUINCHANA AND AGOYA	Sofia Elizabeth				-	Snatch #1
4	GAVIDIA DIAZ	Faviana				-	Snatch #1
5	MARTINEZ ROSALES	Mariangeli				-	Snatch #1
6	PEREIRA RAMOS	Luanne				-	Snatch #1
7	ROSA FIGUEROA	Emily				-	Snatch #1
8	REALPE MINA	Thaira				-	Snatch #1
9	GARCIA SUAREZ	Macarena				-	Snatch #1
10	MORALES AMACIFL	Blanca				-	Snatch #1
11	NOLE PORRAS	Damariz				-	Snatch #1

[3] LIQUINCHANA AND AGOYA, Sofia Elizabeth P15 F 45

snatch 1 snatch 2 snatch 3 C&J 1 C&J 2 C&J 3

Automatic Progression 0 0 0 0

Declaration 45 55

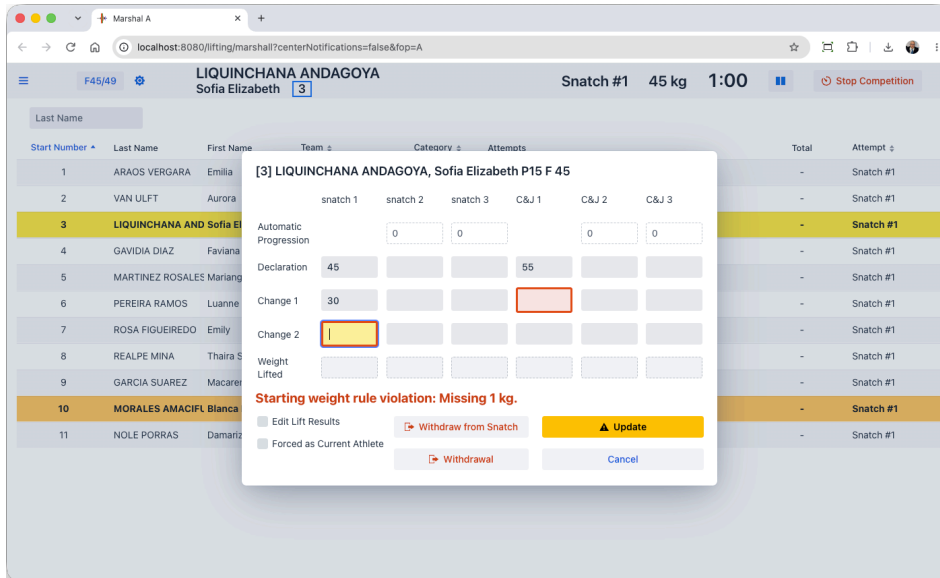
Change 1 1

Change 2

Weight Lifted

☐ Edit Lift Results ☐ Forced as Current Athlete

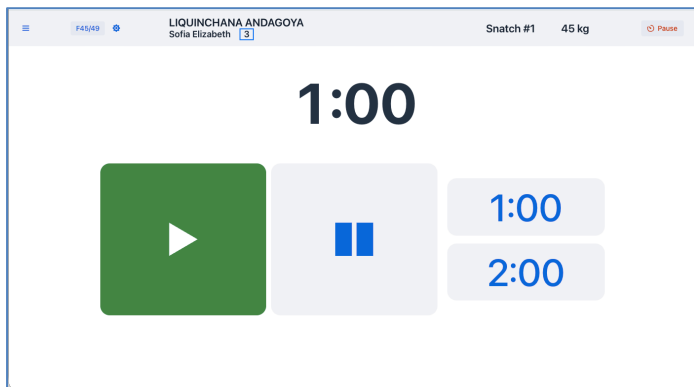
20kg rule violations and late changes are immediately made visible



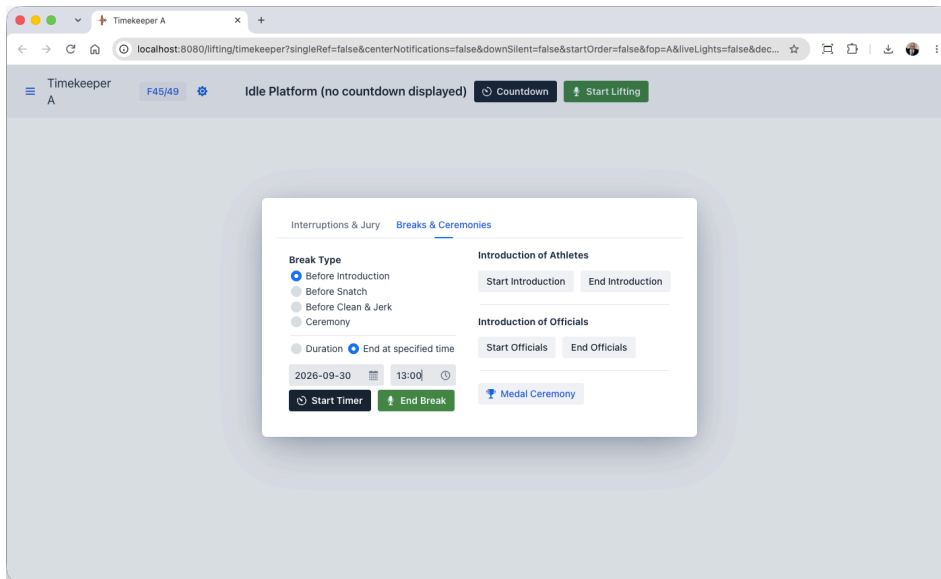
E.2.3.6 Timekeeper

[RFP §4.3] At minimum, the TIS shall provide Timekeeper.

During the competition a tablet-friendly display is used, that can be used together with the Blue Owl buttons, as preferred.



The timekeeper can control the breaks (in addition to the main technical table or the speaker having the same feature depending on the skill and comfort levels). The delay can be set to a clock or duration as relevant for the type of break.



E.2.3.7 Speaker

In OWLCMS, a speaker page shows athletes in lifting order and displays all relevant notifications. This makes it much easier for the speaker, who does not have to interpret in real time what is shown on the scoreboard.

Start Number	Last Name	First Name	Team	Category	Attempts	Total	Attempt
10	MORALES AMACIFUEN	Blanca Mileidy	PER	P15 F 49	45	-	Snatch #1
3	LIQUINCHANA AND	Sofia Elizabeth	ECU	P15 F 45	46	-	Snatch #1
11	NOLE PORRAS	Damaritz Luciana	PER	P15 F 49	50	-	Snatch #1
1	ARAOS VERGARA	Emilia	CHI	P15 F 45	57	-	Snatch #1
4	GAVIDIA DIAZ	Faviana Ayelen	PER	PSR F 49	58	-	Snatch #1
6	PEREIRA RAMOS	Luanne Cristiny	BRA	PSR F 49	72	-	Snatch #1
2	VAN ULFT	Aurora	PER	PSR F 49	72	-	Snatch #1
8	REALPE MINA	Thaira Sara	ECU	PSR F 49	72	-	Snatch #1
9	GARCIA SUAREZ	Macarena Del Pilar	CHI	PSR F 49	68	-	Snatch #1
5	MARTINEZ ROSALES	Mariangeli Irene	VEN	PSR F 49	70	-	Snatch #1
7	ROSA FIGUEROA	Emily	BRA	PSR F 49	72	-	Snatch #1

E.2.3.8 Competition supervisor

[RFP §4.3] At minimum, the TIS shall provide Competition supervisor.

The competition supervisor screen is the as the speaker, with the addition of the decision lights that can be seen live as an option. This is because the switches between sessions, and breaks, and lift overrides after a late referee flag call can all be done from that screen. Depending on competitions, the speaker screen will be made “passive” if the technical table assumes those tasks instead of an ITO1 comfortable with technology.

Start Number	Last Name	First Name	Team	Category	Attempts	Total	Attempt
3	LIQUINCHANA ANDAGOYA	Sofia Elizabeth	ECU	P15 F 45	46	-	Snatch #1
10	MORALES AMACIFUEN, B. M.				45	46	Snatch #2
11	NOLE PORRAS	Damariz			50	-	Snatch #1
1	ARAOS VERGARA	Emilia			57	-	Snatch #1
4	GAVIDIA DIAZ	Faviana		Card	58	-	Snatch #1
6	PEREIRA RAMOS	Luciana			62	-	Snatch #1
2	VAN ULFT	Aurora			65	-	Snatch #1
8	REALPE MINA	Thaira Sarai	ECU	P15 F 49	65	-	Snatch #1
9	GARCIA SUAREZ	Macarena Del Pilar	CHI	PSR F 49	68	-	Snatch #1
5	MARTINEZ ROSALES	Mariangeli Irene	VEN	PSR F 49	70	-	Snatch #1
7	ROSA FIGUEROA	Emily	BRA	PSR F 49	72	-	Snatch #1

In addition, OWLCMS is launched from a control panel that gives additional options such as viewing the technical log going live. The competition supervisor can watch the competition-level audit log as it is written.

Time	Event	Participant	Details
10:13:36:05.916	ANNOUNCER	summary.official	-
11:13:36:05.916	ANNOUNCER	summary.official	-
12:13:36:05.916	ANNOUNCER	summary.official	-
13:13:36:05.916	ANNOUNCER	summary.official	-
14:13:36:05.916	ANNOUNCER	summary.official	-
15:13:36:05.916	ANNOUNCER	summary.official	-
16:13:36:05.916	ANNOUNCER	summary.official	-
17:13:36:05.916	ANNOUNCER	summary.official	-
18:13:36:05.917	ANNOUNCER	summary.athlete	ARAOS VERGARA, Emilia
19:13:36:05.917	ANNOUNCER	summary.athlete	VAN ULFT, Aurora
20:13:36:05.917	ANNOUNCER	summary.athlete	LIQUINCHANA ANDAGOYA, Sofia Elizabeth
21:13:36:05.917	ANNOUNCER	summary.athlete	REALPE MINA, Thaira Sarai
22:13:36:05.917	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
23:13:36:05.917	ANNOUNCER	summary.athlete	NOLE PORRAS, Damariz Luciana
24:13:36:05.917	ANNOUNCER	summary.athlete	GAVIDIA DIAZ, Faviana Avelen
25:13:36:05.917	ANNOUNCER	summary.athlete	MARTINEZ ROSALES, Mariangeli Irene
26:13:36:05.917	ANNOUNCER	summary.athlete	PEREIRA RAMOS, Luciana
27:13:36:05.918	ANNOUNCER	summary.athlete	ROSA FIGUEROA, Emily
28:13:36:05.918	ANNOUNCER	summary.athlete	GARCIA SUAREZ, Macarena Del Pilar
29:13:36:05.918	ANNOUNCER	summary.athlete	LIQUINCHANA ANDAGOYA, Sofia Elizabeth
30:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
31:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
32:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
33:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
34:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
35:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
36:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
37:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
38:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
39:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
40:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
41:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
42:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
43:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
44:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
45:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
46:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
47:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
48:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
49:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
50:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
51:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
52:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
53:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
54:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
55:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
56:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
57:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
58:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
59:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
60:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
61:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
62:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
63:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
64:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
65:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
66:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
67:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
68:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
69:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
70:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
71:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
72:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
73:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
74:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
75:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
76:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy
77:13:36:05.918	ANNOUNCER	summary.athlete	MORALES AMACIFUEN, Blanca Mileidy

E.2.3.9 Referee monitoring

[RFP §4.3] At minimum, the TIS shall provide Referee monitoring.

The competition director screen can see the live referee lights, as does the jury. The audit log shows each individual decision, and each weight change from the marshals.

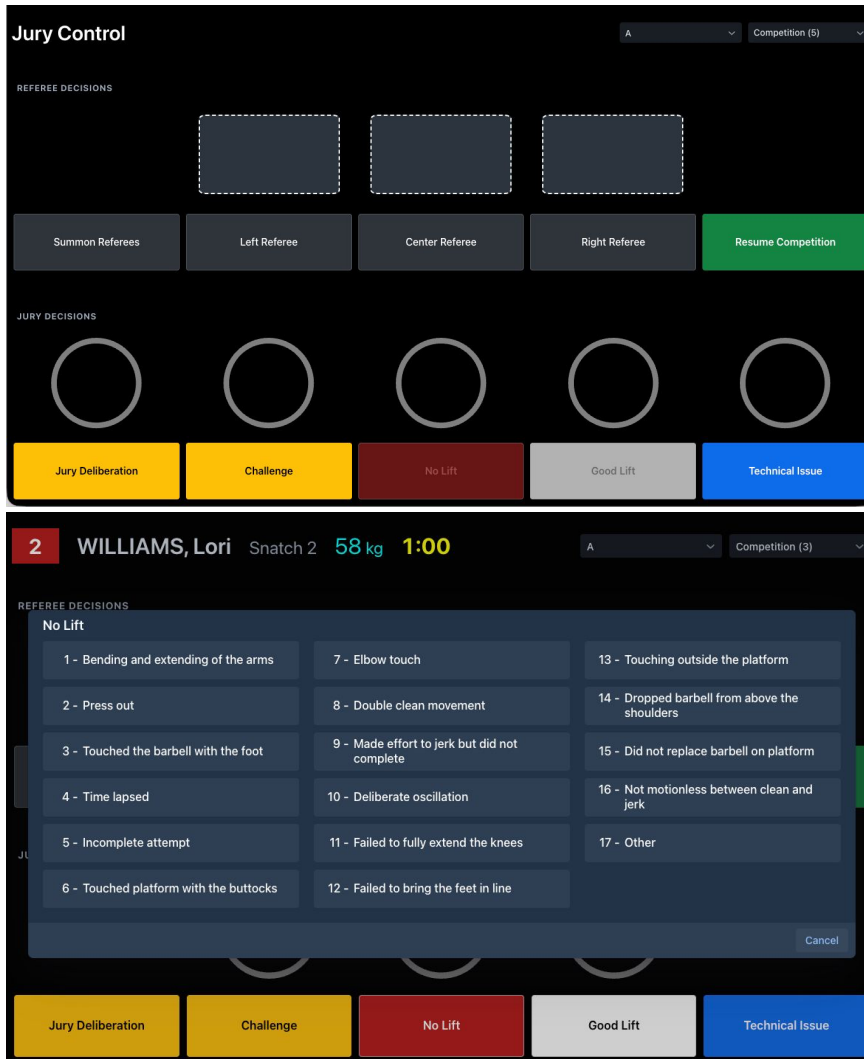
E.2.3.10 Jury control

[RFP §4.3] At minimum, the TIS shall provide Jury control.

The jury control can be done entirely using a Blue Owl jury control keypad



In the context of an IWF meet, it becomes desirable to accelerate the transfer of decisions from the jury to the speaker. To this effect, a tablet-style jury display is available that has the same features as the keypad

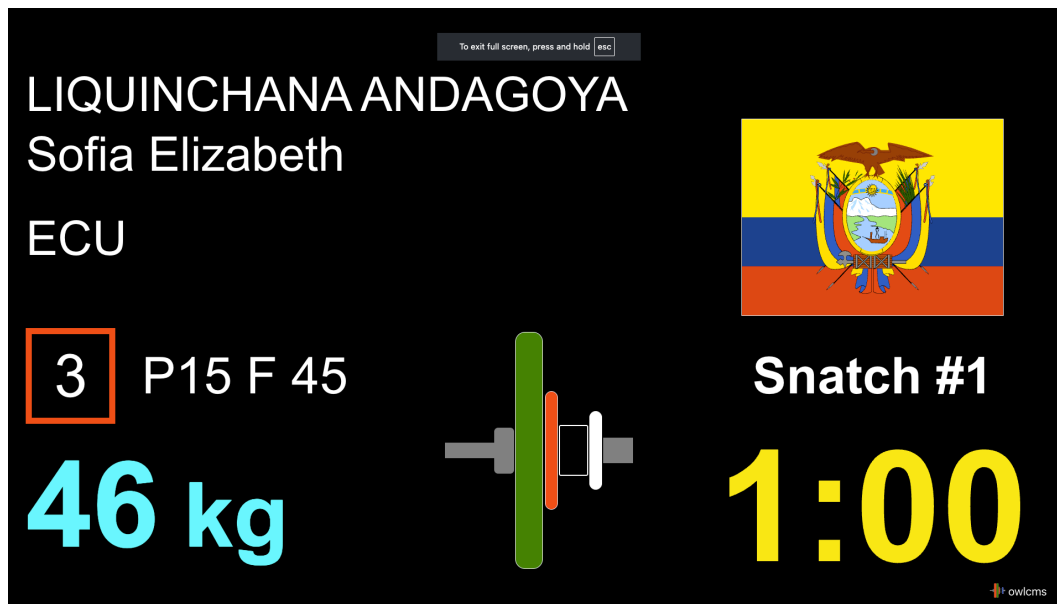


When this is used, the Speaker and Competition Supervisor screens get the reason immediately, and can resume the competition after announcing it.

E.2.3.11 Attempt boards

[RFP §4.3] At minimum, the TIS shall provide Attempt boards.

Attempt boards show all the information required by TCRR. They can also display the flag of the country. If one or more record attempts take place the attempt boards show it as a rolling banner.



E.2.3.12 Scoreboards

[RFP §4.3] At minimum, the TIS shall provide Scoreboards.

Scoreboards contain all the requisite information. In addition, the top line contains the current attempt. The bottom line contains additional information, such as the various championships, rank if successful, jury decisions etc.). This is the warmup room layout, that does not change during ceremonies so coaches can count attempts.

Scoreboard during session with previous leaders

In the main room, styling will be applied as required by the event, and the top row is used as a banner. A row at the bottom carries will carry the clock and other attempt information.

Main Room Scoreboard with previous leaders

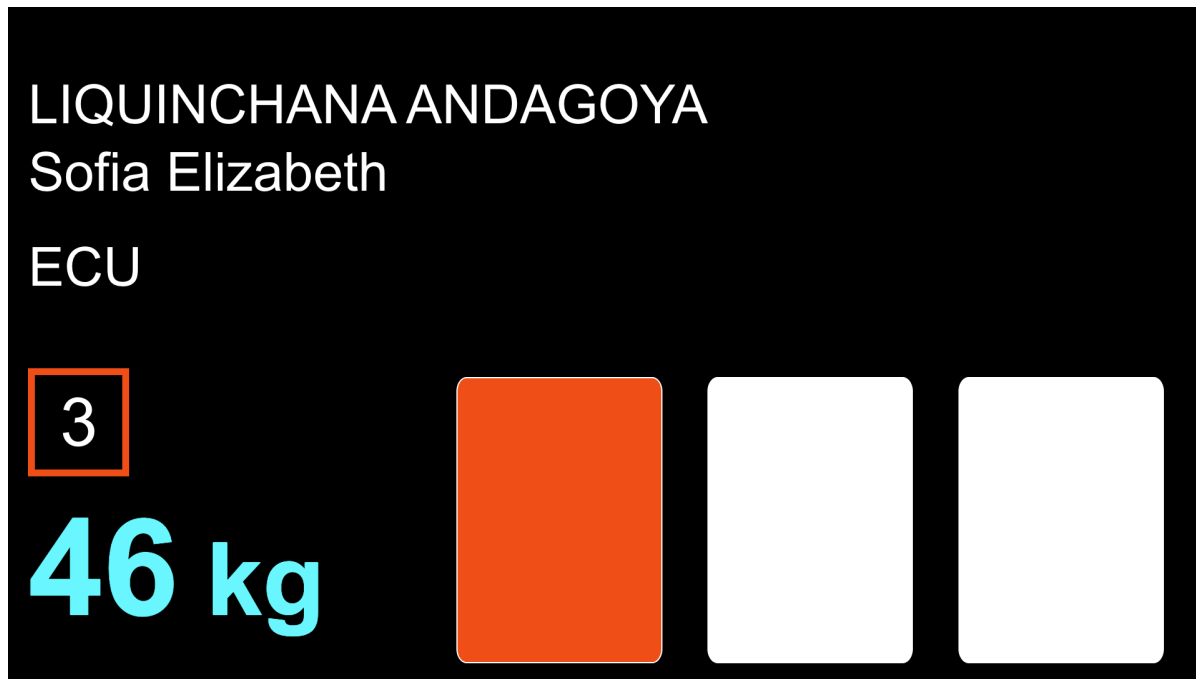
Full-fledged video production companies can replace this scoreboard and the TV Graphics by using the extension capabilities provided by OWLCMS. The extension protocol is documented with a reference implementation at <https://github.com/owlcms/tracker-core>.

E.2.3.13 RTDS/warm-up

[RFP §4.3] At minimum, the TIS shall provide RTDS/warm-up.

In OWLCMS all the screens are RTDS (Real-time Display System)

When decisions are given, they are shown both on the attempt board and on the scoreboards.



If jury decisions are being shown, these appear as well. The bottom line then shows the athlete under review, whereas the top line shows the athlete getting ready (whose clock is not running).

Jury Decision for current athlete

E.2.3.14 STOP

[RFP §4.3] At minimum, the TIS shall provide STOP.

During deliberation, challenge, technical issue or other stoppage, the reason is shown

Flashing Stop

E.2.3.15 Records

[RFP §4.3] At minimum, the TIS shall provide Records.

Record Attempt Attempt Board

Record Attempt Scoreboard

E.2.3.16 Results

[RFP §4.3] At minimum, the TIS shall provide Results.

End of session rankings with medals

E.2.3.17 Other approved states

[RFP §4.3] At minimum, the TIS shall provide Other approved states.

E.2.4 Official outputs

E.2.4.1 Start Lists

[RFP §4.4] At minimum, the TIS shall provide Start Lists.

E.2.4.2 Start List Package

[RFP §4.4] At minimum, the TIS shall provide Start List Package.

E.2.4.3 Weigh-in List

[RFP §4.4] At minimum, the TIS shall provide Weigh-in List.

E.2.4.4 Athlete's Cards

[RFP §4.4] At minimum, the TIS shall provide Athlete's Cards.

E.2.4.5 Empty Protocol

[RFP §4.4] At minimum, the TIS shall provide Empty Protocol.

E.2.4.6 Manual Protocol

[RFP §4.4] At minimum, the TIS shall provide Manual Protocol.

E.2.4.7 Protocol

[RFP §4.4] At minimum, the TIS shall provide Protocol.

E.2.4.8 Results

[RFP §4.4] At minimum, the TIS shall provide Results.

E.2.4.9 Team Classification/Placing

[RFP §4.4] At minimum, the TIS shall provide Team Classification/Placing.

E.2.4.10 Records

[RFP §4.4] At minimum, the TIS shall provide Records.

E.2.4.11 Final Results Package

[RFP §4.4] At minimum, the TIS shall provide Final Results Package.

E.2.4.12 Results Book where applicable

[RFP §4.4] At minimum, the TIS shall provide Results Book where applicable.

E.2.4.13 Machine-readable export

[RFP §4.4] At minimum, the TIS shall provide Machine-readable export.

E.2.4.14 Distribution records

[RFP §4.4] At minimum, the TIS shall provide Distribution records.

E.2.4.15 Final evidence/archive

[RFP §4.4] At minimum, the TIS shall provide Final evidence/archive.

Paper, PDF and machine-readable outputs are delivery channels; none may become an undocumented alternative source of sporting truth.

Supplier response

[TO BE COMPLETED]

AutoSave WeighInForm_Horizontal_2026-09-30_12h13 Search (Cmd + Ctrl + U)

Home Insert Draw Page Layout Formulas Data Review View Automate Comments Share

D3

Name : Pan-American U15, U17 Championship, 1st PanAm Games Qualifier
Site : Guayaquil Ecuador
Date : 2026-08-26

Weigh-In Form

Session F45/49 Weigh-In Time: 2026-08-26 09:00 Start Time: 2026-08-26 11:00

Categoría 45/49 A - Women

Start	Lot	Name	Weight Read on Scale	B.W.	Team	Born	Cat.	Entry	Snatch	Clean&Jerk	Signature
1	41	ARAOS VERGARA, Emilia			CHI	2013	P15 F 45/P17 F 45	137			
2	85	VAN ULFT, Aurora			ARU	2013	P15 F 45/P17 F 45	162			
3	104	LIQUINCHANA ANDAGOYA, Sofia Elizabeth			ECU	2012	P15 F 45/P17 F 45	106			
4	141	REALPE MINA, Thaira Sarai			ECU	2012	P15 F 45/P17 F 45	160			
5	171	MORALES AMACIFUEN, Bianca Mileidy			PER	2011	P15 F 45/P17 F 45	110			
6	179	NOLE PORRAS, Damariz Ludiana			PER	2012	P15 F 45/P17 F 45	120			
7	29	GAVIDIA DIAZ, Faviana Ayelen			PER	2003	PSR F 49	150			
8	72	MARTINEZ ROSALES, Mariangeli Irene			VEN	2003	PSR F 49	176			
9	86	PEREIRA RAMOS, Luanne Crisidy			BRA	2008	PSR F 49	151			
10	137	ROSA FIGUEROA, Emily			BRA	1998	PSR F 49	172			
11	162	GARCIA SUAREZ, Macarena Del Pilar			CHI	2001	PSR F 49	170			

Competition Secretary 1 _____

Competition Secretary 2 _____

Starting Weights

Ready Filter Mode Accessibility: Good to go 100%

Session: F45/49 2026-08-26 11:00 a.m.

Start #

ARAOS VERGARA, Emilia

CHI

Pan-American U15, U17 Championship, 1st PanAm Games Qualifier / Ecuador

Lot	Category	Birth Date
41	P15 F 45	2013

Entry Total **137**

Body Weight Signature:

Snatch

1	2	3
	Automatic Progression	Automatic Progression
Starting Weight	Declaration	Declaration
Signature:	Signature:	Signature:
Change 1	Change 1	Change 1
Signature:	Signature:	Signature:
Change 2	Change 2	Change 2
Signature:	Signature:	Signature:

Session: F45/49 2026-08-26 11:00 a.m.

Start #

ARAOS VERGARA, Emilia

CHI

Pan-American U15, U17 Championship, 1st PanAm Games Qualifier / Ecuador

Lot	Age Group	Category
41	P15	45

20kg Rule **117**

Required C&J Total

Clean&Jerk

1	2	3
	Automatic Progression	Automatic Progression
Starting Weight	Declaration	Declaration
Signature:	Signature:	Signature:
Change 1	Change 1	Change 1
Signature:	Signature:	Signature:
Change 2	Change 2	Change 2
Signature:	Signature:	Signature:

F. Security & resilience

Security architecture, access, offline operation, backup, recovery, failover and continuity.

F.1 Security, Resilience and Business Continuity

F.1.1 Security

OWLCMS meets all stated requirements.

F.1.1.1 Least-privilege access by role and device identity.

[RFP §7.1] Least-privilege access by role and device identity.

OWLCMS supports separate logins for every role, and where relevant, separate logins for each role and platform. Devices are identified per platform and authenticated using the MQTT identity of the MQTT interface.

F.1.1.2 Separation of competition, device, public/display, database and management functions where applicable.

[RFP §7.1] Separation of competition, device, public/display, database and management functions where applicable.

The role structure separates the technical official roles, preparation roles, the competition director, and admin functions.

F.1.1.3 Protection of credentials, secrets and data in transit/at rest where required.

[RFP §7.1] Protection of credentials, secrets and data in transit/at rest where required.

All credentials or secrets are stored encrypted and transported in encrypted form. They are never written to log files in clear. Account passwords use PBKDF2 encryption with HMAC-SHA256.

Todo: tracker connection in database needs propping up.

F.1.1.4 Controlled remote support, removable media and patch/deployment processes.

[RFP §7.1] Controlled remote support, removable media and patch/deployment processes.

Security options are in the database or in external configuration files; updating to a fixed version is independent of the options. All future versions of OWLCMS will support the security requirements of IWF.

F.1.1.5 Tamper-resistant audit sufficient to reconstruct material actions.

[RFP §7.1] Tamper-resistant audit sufficient to reconstruct material actions.

The audit log is tamper-resistant and any tampering is detectable (the logger keeps its own copy of the lines and cryptographically signs blocks of lines). The audit logs includes every single modification to the athlete and lifting data, and all events coming from devices (decisions etc.)

F.1.1.6 No public or anonymous path may write authoritative competition state.

[RFP §7.1] No public or anonymous path may write authoritative competition state.

The only writing to the database is done by authenticated users. The database is the only source of written archivals, that are also cryptographically signed.

F.1.2 Offline operation

[RFP §7.2] Core competition, officiating, timing, local displays and local official documents shall remain usable during loss of public internet or cloud connectivity.

Core competition, officiating, timing, local displays and local official documents shall remain usable during loss of public internet or cloud connectivity.

The normal mode of operation of OWLCMS is completely off-line, on a private local area network, with all key components wired (in particular the refereeing functions).

F.1.3 Backup and recovery

F.1.3.1 Protected backup copies of data, configuration, rules, assets and audit.

[RFP §7.3] Protected backup copies of data, configuration, rules, assets and audit.

All files that constitute the OWLCMS state are in a single folder (with the exception of cryptographic key) Any means of backup can be used – normally copying the folder to a removable device or off-site storage.

F.1.3.2 Timed restoration demonstrating the approved recovery objective.

[RFP §7.3] Timed restoration demonstrating the approved recovery objective.

Restoration of a saved state from scratch normally takes less than a minute.

F.1.3.3 Defined promotion, split-brain prevention, failback and reconciliation where redundancy is claimed.

[RFP §7.3] Defined promotion, split-brain prevention, failback and reconciliation where redundancy is claimed.

The system would not operate in tandem. Checkpoint copies would be used.

F.1.3.4 Cold-start and rollback procedures using controlled recovery media.

[RFP §7.3] Cold-start and rollback procedures using controlled recovery media.

Checkpoint copies of the export format can be obtained on demand, and constitute full rollback points. Restarting the system without a database file creates a minimal empty one that can be immediately loaded from a checkpoint.

F.1.4 Manual continuity

[RFP §7.4] A manual Protocol and rehearsed manual-continuity procedure shall remain available. Staff shall know when to continue, when to STOP and how to reconcile manual records with the electronic state.

A manual Protocol and rehearsed manual-continuity procedure shall remain available. Staff shall know when to continue, when to STOP and how to reconcile manual records with the electronic state.

This is standard operational procedure. A power outage, or a networking mishap can prevent use the system and cause a revert to a “manual cards and flags” mode.

OWLCMS has been tested in such situations before. As soon as the system is restored, someone can simply copy the information as it stands on the cards, even while lifting is going on that platform. As soon as the information has been updated and matches the paper trail, the screens can be turned on for that platform. All the lifting order and such is derived from the authoritative card data as entered.

Writing operational procedures as expected by IWF would take place before deployment.

F.1.5 Monitoring

[RFP §7.5] Actionable monitoring shall cover critical services, devices, timing, storage, network and state-state conditions, with clear escalation responsibilities.

Actionable monitoring shall cover critical services, devices, timing, storage, network and state-state conditions, with clear escalation responsibilities.

The actual OTIS kit requires a router and, for each platform, a switch. As part of the certified kit, this constitutes the monitoring boundary. The kit devices are attached directly to these and can be monitored (including remote logins).

G. Integration

Interfaces, APIs/feeds, public/broadcast and Games/ORIS integrations.

G.1 Interfaces

[RFP §6.3] Material interfaces shall document producer/consumer, purpose, version, schema/field mapping, identifiers, status/null rules, sequencing/revision, time source, latency, authentication, acknowledgement, retry/reconnect, monitoring and fallback.

Material interfaces shall document producer/consumer, purpose, version, schema/field mapping, identifiers, status/null rules, sequencing/revision, time source, latency, authentication, acknowledgement, retry/reconnect, monitoring and fallback.

G.1.1 Device Protocol and Authentication

All the devices on the network communicate using MQTT messages to the OWLCMS server; see the specification online <https://jflamy.github.io/owlcms4/#/100-AdvancedTopics/200-MQTTMessages>

Standard MQTT authentication protocols are used (login name and password); crypted MQTT is available, but this is normally only used on cloud-based configurations, which is not the case for the proposed completely-local, closed-network configuration.

G.1.2 Blue Owl USB-MQTT Interface

- Blue Owl devices use the same computing chip that is used on Arduino Uno or Nano, or on Arduino Mega
- The Blue Owl devices use a standard USB 2.0 interface to communicate with the computer that powers them.
- All communications with the computer are done serially on the same USB connection, using the publicly documented Firmata protocol.
- The Blue Owl program speaks Firmata to the devices and relays the information to OWLCMS
 - When a button is pressed on the device, this is immediately propagated by Blue Owl as an MQTT message to OWLCMS
 - Conversely, when OWLCMS issues a command to turn on an LED or buzzer, the Firmata message activates the corresponding ports.
 - Blue Owl itself will attempt reconnections every second in case of disconnection
 - The server detects and reports when a Blue Owl module disconnects and reconnects
 - Each Blue Owl interface keeps complete logs.
- The actual implementation of the software code used by the Blue Owl user interface to perform the interactions is publicly available at <https://github.com/nemikor-solutions/blue-owl-core>
- Schematics of the Blue Owl devices are included in Annex A.

G.1.3 Direct MQTT Networking

Because the MQTT protocol is documented, any device that can be designed to listen to MQTT can just subscribe to the topics. The VPT modules uses the MQTT messages to know when the clock is started and stopped, as an example. WiFi devices can also act as devices by talking directly to MQTT.

G.1.4 Latency

Latency between the button press and reception of the MQTT message is under XX ms.

H. Operations

Staffing, training, manuals, event support, escalation and incident response.

H.1 Supplier responsibilities

H.1.1 Provide accurate product and hardware baseline

[RFP §8.2] Provide accurate product and hardware baseline.

[To be completed]

H.1.2 Provide complete technical submission and evidence package

[RFP §8.2] Provide complete technical submission and evidence package.

[To be completed]

H.1.3 Make the declared system available for IWF testing and demonstrations

[RFP §8.2] Make the declared system available for IWF testing and demonstrations.

[To be completed]

H.1.4 Correct deficiencies and complete regression testing when required

[RFP §8.2] Correct deficiencies and complete regression testing when required.

[To be completed]

H.1.5 Notify IWF of material product, security, hardware, interface or rule-logic changes

[RFP §8.2] Notify IWF of material product, security, hardware, interface or rule-logic changes.

[To be completed]

H.1.6 Maintain qualified support personnel and appropriate spares

[RFP §8.2] Maintain qualified support personnel and appropriate spares.

[To be completed]

H.1.7 Protect IWF, athlete and event information and comply with applicable data-protection requirements.

[RFP §8.2] Protect IWF, athlete and event information and comply with applicable data-protection requirements.

[To be completed]

H.2 Event deployment

H.2.1 Only the approved baseline and event configuration may be used

[RFP §12.3] Only the approved baseline and event configuration may be used.

[To be completed]

H.2.2 Actual hardware, software, network, power, displays, spares and staffing must match the approved event design.

[RFP §12.3] Actual hardware, software, network, power, displays, spares and staffing must match the approved event design.

[To be completed]

H.2.3 Dress rehearsal and failure/recovery test shall be completed before readiness sign-off.

[RFP §12.3] Dress rehearsal and failure/recovery test shall be completed before readiness sign-off.

[To be completed]

H.2.4 Substitutions require approved deviation or delta testing before use

[RFP §12.3] Substitutions require approved deviation or delta testing before use.

[To be completed]

H.3 Support and incidents

H.3.1 Defined escalation contacts and event support

[RFP §12.4] Defined escalation contacts and event support.

[To be completed]

H.3.2 Critical incidents documented, contained, investigated and reported

[RFP §12.4] Critical incidents documented, contained, investigated and reported.

[To be completed]

H.3.3 Serious incidents, unauthorised changes, false evidence, unsupported baselines or material security compromise may lead to restriction, suspension or withdrawal.

[RFP §12.4] Serious incidents, unauthorised changes, false evidence, unsupported baselines or material security compromise may lead to restriction, suspension or withdrawal.

[To be completed]

I. Homologation

Test environment, evidence package, demonstration plan and FAT/SAT readiness.

I.1 Evaluation and Homologation Process

I.1.1 Administrative review

[RFP §10] Completeness, eligibility and declarations.

[To be completed]

I.1.2 Technical review

[RFP §10] Architecture, functionality, hardware, interfaces, security and resilience.

[To be completed]

I.1.3 Scope confirmation

[RFP §10] Freeze baseline, profile, capacity, dependencies and test plan.

[To be completed]

I.1.4 FAT

[RFP §10] Controlled normal, edge, security, capacity and failure testing.

[To be completed]

I.1.5 Corrective action

[RFP §10] Resolve deficiencies and provide regression evidence.

[To be completed]

I.1.6 Operational validation

[RFP §10] Representative event tempo with operators, officials, displays, feeds and recovery.

[To be completed]

I.1.7 Homologation decision

[RFP §10] Review requirements, hard gates, evidence and findings.

[To be completed]

I.1.8 Supplier licence

[RFP §10] Complete legal and operational conditions.

[To be completed]

I.1.9 Event SAT

[RFP §10] Check actual installation against approved baseline.

[To be completed]

I.1.10 Surveillance

[RFP §10] Review incidents, changes, support and expiry.

[To be completed]

I.1.11 Evidence

[RFP §10.1] Evidence shall be attributable to a specific baseline and test. Screenshots alone are insufficient where logs, physical inspection or live behaviour are necessary.

Evidence shall be attributable to a specific baseline and test. Screenshots alone are insufficient where logs, physical inspection or live behaviour are necessary.

Supplier response

[TO BE COMPLETED]

I.2 Approval Criteria and Hard Gates

I.2.1 HG-01 Sport rules & result correctness

[RFP §11] Incorrect sporting fact, rank, medal, record or rule-critical transition.

[To be completed]

I.2.2 HG-02 Authoritative state & data integrity

[RFP §11] Split, duplicated, lost or silently overwritten accepted state.

[To be completed]

I.2.3 HG-03 Timing & adjudication safety

[RFP §11] Unsafe or ambiguous timing, decision, Down signal or restart.

[To be completed]

I.2.4 HG-04 Correction & audit

[RFP §11] Unauthorised, partial, untraceable or unreconciled correction.

[To be completed]

I.2.5 HG-05 Continuity & recovery

[RFP §11] Uncontrolled loss of competition or unverified recovery.

[To be completed]

I.2.6 HG-06 Security & least privilege

[RFP §11] Uncontrolled privilege, public write path or material compromise.

[To be completed]

I.2.7 HG-07 Output parity & publication

[RFP §11] Contradictory outputs or untraceable final publication.

[To be completed]

I.2.8 HG-08 Baseline reproducibility

[RFP §11] Unknown tested version, missing evidence or irreproducible installation.

[To be completed]

I.2.9 HG-09 FOP readiness & safety

[RFP §11] Missing critical equipment, unsafe installation or unresolved obstruction.

[To be completed]

I.2.10 Decision rules

I.2.10.1 All applicable mandatory requirements and tests must be disposed of

[RFP §11.1] All applicable mandatory requirements and tests must be disposed of.

[To be completed]

I.2.10.2 Applicable TCRR requirements and hard gates must pass

[RFP §11.1] Applicable TCRR requirements and hard gates must pass.

[To be completed]

I.2.10.3 No open Critical finding may remain

[RFP §11.1] No open Critical finding may remain.

[To be completed]

I.2.10.4 Traceability from requirement to evidence and test must be complete

[RFP §11.1] Traceability from requirement to evidence and test must be complete.

[To be completed]

I.2.10.5 Successful FAT, recovery and operational validation are required for the claimed profile.

[RFP §11.1] Successful FAT, recovery and operational validation are required for the claimed profile.

[To be completed]

I.2.10.6 Partial is not Pass

[RFP §11.1] Partial is not Pass.

[To be completed]

I.2.10.7 A profile limitation cannot change an IWF sport rule or bypass a hard gate

[RFP §11.1] A profile limitation cannot change an IWF sport rule or bypass a hard gate.

[To be completed]

J. Commercial

Licence, implementation, equipment, support/maintenance, optional services and recurring charges.

J.1 Commercial and Contractual Conditions

The Licensing Fee for the four-year period is US\$20,000, payable on an annual basis of US\$5,000 per year.

J.1.1 IWF licensing fee

[RFP §13] The Licensing Fee for the four-year period is US\$20,000, payable on an annual basis of US\$5,000 per year.

US\$5,000 per year / US\$20,000 total

Assumptions: [TO BE COMPLETED]

J.1.2 Licence

[RFP §13] Licence

[TO BE COMPLETED]

Assumptions: [TO BE COMPLETED]

J.1.3 Implementation

[RFP §13] Implementation

[TO BE COMPLETED]

Assumptions: [TO BE COMPLETED]

J.1.4 Equipment

[RFP §13] Equipment

[TO BE COMPLETED]

Assumptions: [TO BE COMPLETED]

J.1.5 Support / maintenance

[RFP §13] Support / maintenance

[TO BE COMPLETED]

Assumptions: [TO BE COMPLETED]

J.1.6 Optional services

[RFP §13] Optional services

[TO BE COMPLETED]

Assumptions: [TO BE COMPLETED]

J.1.7 Recurring charges

[RFP §13] Recurring charges

[TO BE COMPLETED]

Assumptions: [TO BE COMPLETED]

K. Legal

IP/licensing, warranties, liability, confidentiality, data protection, subcontractors and termination.

K.1 Licence Conditions, Change Control and Event Deployment

K.1.1 Licence validity

[RFP §12.1] The IWF licence shall be valid for a four-year period. The certificate shall state supplier, product/baseline, profiles, capacities, limitations, deviations, validity and surveillance conditions.

The IWF licence shall be valid for a four-year period. The final contract shall state the exact licence duration and renewal conditions.

Supplier response

[TO BE COMPLETED]

K.1.2 Change control

K.1.2.1 Editorial / non-functional

[RFP §12.2] Controlled update and review.

[TO BE COMPLETED]

K.1.2.2 Minor cosmetic

[RFP §12.2] Supplier assessment and IWF notification where applicable.

[TO BE COMPLETED]

K.1.2.3 Significant - OS/database/firmware/interface/security/workflow/hardware

[RFP §12.2] Impact assessment and approved regression/delta test.

[TO BE COMPLETED]

K.1.2.4 Major - competition logic/timing/adjudication/records/state/failover

[RFP §12.2] Re-homologation before event use.

[TO BE COMPLETED]

K.1.2.5 Emergency security/stability correction

[RFP §12.2] Named approval, backup, rollback, targeted test and retrospective evidence.

[TO BE COMPLETED]

K.2 Intellectual property

[RFP §12.5] IWF shall receive the rights, documentation, exports, interfaces and operational access necessary to use, verify, archive and maintain the approved system and competition records.

The supplier retains its intellectual property except where otherwise agreed. IWF shall receive the rights, documentation, exports, interfaces and operational access necessary to use, verify, archive and maintain the approved system and competition records. Final contractual wording will be agreed with the selected supplier.

Supplier response

[TO BE COMPLETED]

K.3 Mandatory declarations

K.3.1 Acceptance of the IWF TCRR as the controlling sporting rules

[RFP §9.1] Acceptance of the IWF TCRR as the controlling sporting rules.

[TO BE COMPLETED]

K.3.2 Identification of all material limitations and dependencies

[RFP §9.1] Identification of all material limitations and dependencies.

[TO BE COMPLETED]

K.3.3 Ability to reproduce the tested system from controlled installation material

[RFP §9.1] Ability to reproduce the tested system from controlled installation material.

[TO BE COMPLETED]

K.3.4 Offline operation and tested recovery for the claimed profile

[RFP §9.1] Offline operation and tested recovery for the claimed profile.

[TO BE COMPLETED]

K.3.5 No automatic licensing mechanism that can interrupt an active competition

[RFP §9.1] No automatic licensing mechanism that can interrupt an active competition.

[TO BE COMPLETED]

Authorised signatory	Details
Name	[TO BE COMPLETED]
Title	[TO BE COMPLETED]
Signature	[TO BE COMPLETED]
Date	[TO BE COMPLETED]

L. Compliance matrix

Clear response to this tender and the applicable IWF assessment.

RFP Section 2

Requirement	Compliance	Reference	Justification
2.1 Complete sport logic, mandatory FOP controls, screens and official documents.	Compliant	B.2.1	OWLCMS and Blue Owl are fully compliant with the TCRR as of October 2026.
2.1-R02 M0 plus baseline control, audit, interfaces, offline operation, recovery, procedures, training, support and spares.	Compliant	B.2.2	The system as delivered will meet all these requirements. The Tracker element illustrated above provides a fully documented interface for all complementary systems that require one in more advanced profiles.
2.1-R03 M1 plus applicable VPT, public/live services, broadcast outputs, enhanced monitoring, high availability and full rehearsal.	Compliant	B.2.3	Claimed: Yes. Video playback requires acquisition of additional video-capable equipment (laptops with video capture/gaming cards, cameras). OWLCMS has its own standalone VPT module that would be covered by this proposal. OWLCMS provides a set of TV Graphics for broadcasts, and its replay streams can be used as well. Supplier routinely does event-specific TV Graphics and has used the OWLCMS integration features to do so at large multisport games.
2.1-R04 M2 plus approved Games/ORIS and programme-specific integrations.	Compliant	B.2.4	Claimed: Yes. Supplier has previously integrated OWLCMS with the ODF (Olympic Data Format) for Games integration in the past (Islamic Solidarity Games 2025). The Tracker module uses the exact same protocol that is available for any external integration. Full access to the competition data and to all competition events is provided over a secure connection, with TLS encryption (wss) supported. Figure 1: Solution Overview Figure 2: Detail: VPT and Interfaces
2.1-R05 Simultaneous platform operation with separation and capacity evidence.	Compliant	B.2.5	Claimed: Yes. OWLCMS has supported up to 6 platforms simultaneously with extremely low computer and network usage and 1600 athletes competing. The program is verified using a stress test designed explicitly for this purpose. When used in this mode, a single instance of OWLCMS runs all platforms, and no integration step for results is required. TCRR mandates that a lower entry group (e.g., group B) lifts before a higher one (e.g. group A); if this constraint is observed, all live results are correct.
2.2 The licence certificate shall state the supplier and product.	Compliant	B.3.1	The supplier of the product shall be [Supplier], who will act as integrator of the open-source OWLCMS version, of the Blue Owl OTIS kit to profile MP, and of the necessary additional components required to meet profile M3.
2.2-R02 The licence certificate shall state the exact baseline.	Compliant	B.3.2	Any stable version of OWLCMS after version 69 is claimed to meet the current requirements and is fully compliant with TCRR as published at the time of this tender. Any version of the Blue Owl software after version 3.0 meets the

Requirement	Compliance	Reference	Justification
			requirements. Final baseline will be confirmed after acceptance testing.
2.2-R03 The licence certificate shall state approved profiles and capacities.	Compliant	B.3.3	M2 and MP profiles: The system has been tested to 6 concurrent platforms and 1600 participants, and can be stress-tested on demand. Examples of such very large mixed events include USAW Open 2024, USAW VWS Series 2025 and 2026, IMWA World Championships M3 profile: [Supplier] integrated the OVR into the Islamic Solidarity Games 2025 central platform using ODF (Olympic Data Format) protocols. [Supplier] will provide interfaces for the protocols used at given major events.
2.2-R04 The licence certificate shall state interfaces and critical dependencies.	Compliant	B.3.4	The system has no critical dependencies. All communications use the standard HTTP and MQTT protocols. The specific MQTT messages used for the devices are fully documented. The protocol used to create additional interfaces for Data Exports, TV Graphics, etc. is public and has a published reference implementation.
2.2-R05 The licence certificate shall state limitations and deviations.	Compliant	B.3.5	None known.
2.2-R06 The licence certificate shall state validity and surveillance conditions.	Compliant	B.3.6	Licensing authorizes the use of the TIS equipment and software package; the equipment and software may not be used if the user does not hold such a license. IWF is expected to acquire TIS equipment and software licences from [Supplier] for IWF use at competitions that are directly under IWF authority. A Federation licensing the TIS equipment and software package is expected to use it for the competitions that are under its own authority. If the Federation wishes to lend the equipment to another federation, or to a constituent federation (such as a national federation loaning the equipment to a state federation), additional licensing fees may be required by the Supplier.

RFP Section 3

Requirement	Compliance	Reference	Justification
3-R01 Import and validate athlete, federation/NOC, official, event, venue and record data; stable IDs and error reporting.	Compliant	E.1.1	Data is converted to an Excel-style sheet with the athlete identifier, name, and all registration data. Competition information and dates are stored in the program database (all information is stored in a relational database). The initial schedule is part of the data intake, and an initial assignment of athletes to sessions is done. The input format handles multiple age groups and supports team allocation. The intake format can be exported, edited off-line, and re-ingested without loss of information. The same applies to schedule changes and referee assignments that can be modified in bulk until stabilized.
3-R02 Corrections, approvals, random lots, grouping, timetable and Technical Official assignments.	Compliant	E.1.2	The program prints VFE verification sheets that show exactly which athlete is in which category or categories. There are also team membership teams to verify that all team allocations are correct. These sheets can be signed by team representatives to facilitate the VFE process and reduce the number of errors, or used by the TIS team to do the capture and then get them signed.
3-R03 Approved Start List and controlled distribution/versioning.	Compliant	E.1.3	OWLCMS as proposed is able to produce a fully compliant start book. Preliminary versions can be published to list participants, with subsequent editions adding the schedule and the officials assignments. Once converted to the OWLCMS format, any number of records can be handled. Field-of-play formats are also available (actual session assignments and schedule) for direct use by coaches and technical officials.
3-R04 Correct timing, bodyweight, first attempts, 20 kg checks, start numbers, signatures and closed snapshot.	Compliant	E.1.4	OWLCMS produces field-tested Weigh-in forms to capture both the weight-on-scale and deducted weight. All applicable rule are checked, including 20 kg and qualifying totals where applicable. The form captures all the required signature. Compliant athlete cards are likewise printed. A special data entry mode is used for quick capture of weigh-in data. A post-weigh-in snapshot of the database is taken. Start numbers are cross-checked at the end of the weigh-in and checked against allocated bib numbers. A speaker introduction sheet for athletes and officials is produced, as are the empty protocols and jury member sheets.
3-R05 Calling order, declarations/changes, 60/120-second timing, referee decisions, records and live outputs.	Compliant	E.1.5	Full compliance with the competition flow is attained. All marshal data entry is immediately handled by the system and all lifting order or requested weights are instantly updated on all displays and scoreboards. All the normal timing decisions are automatic (the clock is automatically set to the right allocated time for each athlete, including on fast sequences of changes). Only explicit loading incidents require a reset. All attempts are checked against the 20 kg rule and the required changes are explicitly visible to the marshal. Incomplete situations that do not fully restore the 20kg rule are explicitly visible.
3-R06 Controlled Snatch-to-Clean & Jerk	Compliant	E.1.6	Default rules for longer breaks (few athletes) and shorter breaks (large sessions) are both available. The session

Requirement	Compliance	Reference	Justification
transition and synchronized displays.			schedule also allows changing the break duration. This can also be done after the session has started (under the assumption the meet director informs all concerned). Transition from the snatch to the clean & jerk break at the end of the snatch is automatic. A jury interruption or challenge on the last requires the time keeper starts the break once the situation has been resolved.
3-R07 Calling order, weight changes, 20 kg enforcement, totals, ranks and records.	Compliant	E.1.7	The same completely automatic timing and weight change flow as in the snatch takes place.
3-R08 STOP, Jury, challenge/VPT where applicable, additional attempts, correction, withdrawal and restart.	Compliant	E.1.8	The jury can start a deliberation or accept a challenge using a physical jury keypad button or using a jury tablet. A flashing STOP sign is shown to the athlete on the platform, and all attempt boards and displays indicate the interruption. Transmission of the jury reversal code to the speaker can be done the traditional way (by the technical controller), or automatically when the jury president chooses the reversal code. In either case, the jury or challenge decision is shown as the speaker announces it. The system can optionally display jury decisions on the scoreboards and the attempt boards. This capability is optional because it is not formally described in TCRR.
3-R09 Protocol reconciliation, signatures/status, results, classifications and records.	Compliant	E.1.9	A session protocol sheet is available immediately after the session that contains all the requisite information and each occurrence of an improved record. This is normally signed by the jury president. After a session, any correction required can be made via the system to ensure an audit trail to match whatever decisions were written on the protocols.
3-R10 Medallists, public/broadcast outputs, communications and revised outputs.	Compliant	E.1.10	The system distinguishes two types of scoreboards. Public scoreboards are normally styled to match the event colors, and switch content during ceremonies. OWLCMS will automatically switch during introductions. During medal ceremonies, the technical table (or the speaker themselves) can change the display to show the medals. In multi-age or multi-federation championships, this allows correct control of what is shown to the public. If broadcasters elect to use the video layout screens as provided by OWLCMS, the changes are also automatic.
3-R11 Final dataset, official outputs, logs, revisions, evidence and machine-readable export.	Compliant	E.1.11	The system distinguishes two types of scoreboards. Public scoreboards are normally styled to match the event colors, and switch content during ceremonies. OWLCMS will automatically switch during introductions. During medal ceremonies, the technical table (or the speaker themselves) can change the display to show the medals. In multi-age or multi-federation championships, this allows correct control of what is shown to the public. If broadcasters elect to use the video layout screens as provided by OWLCMS, the changes are also automatic.

Requirement	Compliance	Reference	Justification
3.1-R01 Snatch followed by Clean & Jerk, maximum three attempts per lift, correct Total calculation.	Compliant	E.1.12.1	Fully compliant with TCRR Evidence / reference: Explicitly in database schema, automated tests.
3.1-R02 Whole-kilogram weights and applicable minimum starting weights.	Compliant	E.1.12.2	Fully compliant with TCRR Evidence / reference: Explicitly in database schema, automated tests.
3.1-R03 Correct calling order, including weight, attempt number, previous-attempt sequence and start number.	Compliant	E.1.12.3	Fully compliant with TCRR Evidence / reference: Automated tests for all the change sequences.
3.1-R04 Permitted weight changes, final-call and consecutive-attempt rules.	Compliant	E.1.12.4	Fully compliant with TCRR Evidence / reference: Automated tests for all the change sequences and for the two-minute situations. Systematic recalculation of lift order to detect illegal moving down, with automatic tests.
3.1-R05 Withdrawals, no-lift/no-Total conditions, disqualification and additional attempts.	Compliant	E.1.12.5	Fully compliant with TCRR Evidence / reference: Automated tests of the total, disqualification and withdrawals.
3.1-R06 Material corrections recalculate affected rankings, records and outputs.	Compliant	E.1.12.6	Fully compliant with TCRR Evidence / reference: Any change to the lifting data causes a recalculation of all affected items. Automated tests for the ranking and records calculations.

RFP Section 4

Requirement	Compliance	Reference	Justification
4.1-R01 Authoritative electronic countdown in one-second increments, configurable up to at least 15 minutes.	Compliant	E.2.1.1	Fully compliant with TCRR Evidence / reference: Timekeeper controls all break durations, no limit on duration.
4.1-R02 Automatic 60- and 120-second attempt clocks and applicable introduction/break countdowns.	Compliant	E.2.1.2	Fully compliant with TCRR Evidence / reference: Clock ownership is automatic, no intervention required to set the clock to 1:00, 2:00 or to the time remaining for the athlete. Transition between countdowns (introduction to snatch for example) is automatic. Can be overridden as required.
4.1-R03 Correct start, stop, restart and STOP behaviour according to the rules.	Compliant	E.2.1.3	Fully compliant with TCRR Evidence / reference: As per TCRR
4.1-R04 All connected displays use the same authoritative time state within the approved tolerance.	Compliant	E.2.1.4	Fully compliant with TCRR Evidence / reference: Time is kept on the OWLCMS server; when sent to remote devices, actual universal time reference for timer termination so the device can compensate for eventual small clock drift. All devices are time-synchronized.
4.1-R05 Time corrections are authorised, recorded and auditable.	Compliant	E.2.1.5	Fully compliant with TCRR Evidence / reference: The displays are NTP-synchronized at the operating system level by default. Verifying time synchronization is part of the technical setup.
4.2-R01 Three independent referee control boxes with white/red decisions and required indicators.	Compliant	E.2.2.1	Fully compliant with TCRR Evidence / reference: Nemikor referee buttons have visual indicator, sound, and vibration.
4.2-R02 Correct majority/Down signal, reminder and reversal-window behaviour.	Compliant	E.2.2.2	Fully compliant with TCRR Evidence / reference: Implemented by software. Reversal rules are as per TCRR. Manual reversal of a decision following flags can easily be done by the technical table.
4.2-R03 Jury visibility of individual referee decisions before public reveal.	Compliant	E.2.2.3	Fully compliant with TCRR Evidence / reference: Live referee decisions and reversal are shown to the jury both on the keypad and on the jury console screen or tablet display if used.
4.2-R04 Support for approved three- and five-member Jury configurations.	Compliant	E.2.2.4	Fully compliant with TCRR Evidence / reference: Jury size is selectable,
4.2-R05 Individual Jury decision controls and protected	Compliant	E.2.2.5	Fully compliant with TCRR Evidence / reference: Individual jury buttons are provided as per TCRR. Decision protection housings are not a TCRR requirement but will be addressed.

Requirement	Compliance	Reference	Justification
independence between members.			
4.2-R06 Material decisions and actions are attributable and time-stamped.	Compliant	E.2.2.6	Fully compliant with TCRR Evidence / reference: Full audit log of every single decision, timestamped to the millisecond, with the originating station or device.
4.3-R01 At minimum, the TIS shall provide System health.	Compliant	E.2.3.1	A technical log is always available for detailed technical operations. The memory, disk and networking usage are extremely low, the system gauges confirm this (the system never exceeds 20%, and is usually well below 10, even for extremely large meets).
4.3-R02 At minimum, the TIS shall provide Event setup.	Compliant	E.2.3.2	All competition information setup is done using interactive screens. Green is used in the software to indicate common actions.
4.3-R04 At minimum, the TIS shall provide lots/groups/timetable.	Compliant	E.2.3.2.2	Normally an import from a competitions registration system is used to set up the initial schedule and assign athletes assigned to A/B/C... groups. At the scale of IWF events, this is most likely done using output of the registration system in spreadsheet format. This is then converted to fit the documented OWLCMS format (this has now become trivial with human assistance tools). All data exchange formats for OWLCMS are public. Once this is done all athletes can be edited in the program, and late entries added by hand if needed. The registration screen allows modification of the allocated session, moving the athlete to another group, adjusting eligibility to multiple age groups as appropriate. The other tabs on the registration form cover things like record eligibility (in joint jurisdiction championships, not all athletes are eligible to improve all records) The session timetable is editable.
4.3-R07 At minimum, the TIS shall provide Weigh-in.	Compliant	E.2.3.4	The weigh-in data entry form is intentionally the same format as the registration form. All relevant checks for the 20 kg rule are applied at data entry, in case they were missed by the weigh-in Tos. After weigh-in, the page allows the creation of all the relevant forms for the speaker and for the jury.
4.3-R08 At minimum, the TIS shall provide Chief Marshal.	Compliant	E.2.3.5	The marshal table has a station that shows all athletes (in start order or lifting order as preferred by the TO operating). Clicking on an athlete brings up a card that is laid out exactly as the athlete card, and automatically brings focus to the right cell, such that typing and hitting Enter can be used for fast data entry. 20kg rule violations and late changes are immediately made visible
4.3-R09 At minimum, the TIS shall provide Timekeeper.	Compliant	E.2.3.6	During the competition a tablet-friendly display is used, that can be used together with the Blue Owl buttons, as preferred. The timekeeper can control the breaks (in addition to the main technical table or the speaker having the same feature depending on the skill and comfort levels). The

Requirement	Compliance	Reference	Justification
			delay can be set to a clock or duration as relevant for the type of break.
4.3-R11 At minimum, the TIS shall provide Competition supervisor.	Compliant	E.2.3.8	The competition supervisor screen is the as the speaker, with the addition of the decision lights that can be seen live as an option. This is because the switches between sessions, and breaks, and lift overrides after a late referee flag call can all be done from that screen. Depending on competitions, the speaker screen will be made “passive” if the technical table assumes those tasks instead of an ITO1 comfortable with technology. In addition, OWLCMS is launched from a control panel that gives additional options such as viewing the technical log going live. The competition supervisor can watch the competition-level audit log as it is written.
4.3-R12 At minimum, the TIS shall provide Referee monitoring.	Compliant	E.2.3.9	The competition director screen can see the live referee lights, as does the jury. The audit log shows each individual decision, and each weight change from the marshals.
4.3-R13 At minimum, the TIS shall provide Jury control.	Compliant	E.2.3.10	The jury control can be done entirely using a Blue Owl jury control keypad In the context of an IWF meet, it becomes desirable to accelerate the transfer of decisions from the jury to the speaker. To this effect, a tablet-style jury display is available that has the same features as the keypad When this is used, the Speaker and Competition Supervisor screens get the reason immediately, and can resume the competition after announcing it.
4.3-R14 At minimum, the TIS shall provide Attempt boards.	Compliant	E.2.3.11	Attempt boards show all the information required by TCRR. They can also display the flag of the country. If one or more record attempts take place the attempt boards show it as a rolling banner.
4.3-R15 At minimum, the TIS shall provide Scoreboards.	Compliant	E.2.3.12	Scoreboards contain all the requisite information. In addition, the top line contains the current attempt. The bottom line contains additional information, such as the various championships, rank if successful, jury decisions etc.). This is the warmup room layout, that does not change during ceremonies so coaches can count attempts. Scoreboard during session with previous leaders In the main room, styling will be applied as required by the event, and the top row is used as a banner. A row at the bottom carries will carry the clock and other attempt information. Main Room Scoreboard with previous leaders Full-fledged video production companies can replace this scoreboard and the TV Graphics by using the extension capabilities provided by OWLCMS. The extension protocol is documented with a reference implementation at https://github.com/owlcms/tracker-core .
4.3-R16 At minimum, the TIS shall provide RTDS/warm-up.	Compliant	E.2.3.13	In OWLCMS all the screens are RTDS (Real-time Display System) When decisions are given, they are shown both on the attempt board and on the scoreboards. If jury decisions are being shown, these appear as well. The

Requirement	Compliance	Reference	Justification
			bottom line then shows the athlete under review, whereas the top line shows the athlete getting ready (whose clock is not running). Jury Decision for current athlete
4.3-R17 At minimum, the TIS shall provide STOP.	Compliant	E.2.3.14	During deliberation, challenge, technical issue or other stoppage, the reason is shown Flashing Stop
4.3-R18 At minimum, the TIS shall provide Records.	Compliant	E.2.3.15	Record Attempt Attempt Board Record Attempt Scoreboard
4.3-R19 At minimum, the TIS shall provide Results.	Compliant	E.2.3.16	End of session rankings with medals
4.3-R20 At minimum, the TIS shall provide Other approved states.	Compliant	E.2.3.17	[TO BE COMPLETED]
4.4-R01 At minimum, the TIS shall provide Start Lists.	Compliant	E.2.4.1	[TO BE COMPLETED]
4.4-R02 At minimum, the TIS shall provide Start List Package.	Compliant	E.2.4.2	[TO BE COMPLETED]
4.4-R03 At minimum, the TIS shall provide Weigh-in List.	Compliant	E.2.4.3	[TO BE COMPLETED]
4.4-R04 At minimum, the TIS shall provide Athlete's Cards.	Compliant	E.2.4.4	[TO BE COMPLETED]
4.4-R05 At minimum, the TIS shall provide Empty Protocol.	Compliant	E.2.4.5	[TO BE COMPLETED]
4.4-R06 At minimum, the TIS shall provide Manual Protocol.	Compliant	E.2.4.6	[TO BE COMPLETED]
4.4-R07 At minimum, the TIS shall provide Protocol.	Compliant	E.2.4.7	[TO BE COMPLETED]
4.4-R08 At minimum, the TIS shall provide Results.	Compliant	E.2.4.8	[TO BE COMPLETED]
4.4-R09 At minimum, the TIS shall provide Team Classification/Placing.	Compliant	E.2.4.9	[TO BE COMPLETED]
4.4-R10 At minimum, the TIS shall provide Records.	Compliant	E.2.4.10	[TO BE COMPLETED]

Requirement	Compliance	Reference	Justification
4.4-R11 At minimum, the TIS shall provide Final Results Package.	Compliant	E.2.4.11	[TO BE COMPLETED]
4.4-R12 At minimum, the TIS shall provide Results Book where applicable.	Compliant	E.2.4.12	[TO BE COMPLETED]
4.4-R13 At minimum, the TIS shall provide Machine-readable export.	Compliant	E.2.4.13	[TO BE COMPLETED]
4.4-R14 At minimum, the TIS shall provide Distribution records.	Compliant	E.2.4.14	[TO BE COMPLETED]
4.4-R15 At minimum, the TIS shall provide Final evidence/archive.	Compliant	E.2.4.15	Paper, PDF and machine-readable outputs are delivery channels; none may become an undocumented alternative source of sporting truth. Supplier response [TO BE COMPLETED]

RFP Section 5

Requirement	Compliance	Reference	Justification
5-R01 Controlled capacity for authoritative state and approved services, with spare/recovery path.	Compliant	D.1.1	One main control computer with 8GB of memory and a 256GB disk is completely sufficient. ARM64 architecture as embodied in a Raspberry Pi5 is demonstrated to handle 6 platforms with 1600 athletes with competition events taking place as fast as rules allow (stress test) and while providing all the requisite screens and attempt boards. The proposed configuration includes a spare. Furthermore computer or laptop running Windows, Linux or macOS can be substituted; the OWLCMS software runs identically on all these platforms, and the machine-readable database exports are readable on all.
5-R02 Dedicated, role-controlled competition administration and operation positions.	Compliant	D.1.2	Operator stations are only required to run one or two browsers and, for some, the Blue Owl software when devices are connected. As a consequence, any machine with 4GB or more of memory can be used. The proposed configuration uses 8GB Raspberry Pi500 that are easily replaceable, will be available as such for 15 years, and are designed to be used in difficult environments such as classrooms. Keyboards with numeric pads will be provided for the Marshal and Data entry stations.
5-R03 Protected authoritative storage, backup and tested recovery.	Compliant	D.1.3	A fully relational database called H2 is used. The volume of data used even in a very large competition is quite small, and H2 is extremely quick. It has been used without a single attributable failure since 2009. The actual H2 binary file can be kept, in addition to the fully machine readable exports. The readable export is the primary means of backup and snapshot. A readable snapshot can be obtained at any time from the administration or competition director stations, and can also be configured to be obtained on other stations using the tracker protocol.
5-R04 Three independent red/white referee control devices.	Compliant	D.1.4	The Blue Owl referee controls include the two buttons, a visible LED, a sound, and a vibration for warning the referees. Referee controllers are cabled in order to avoid any concern with batteries.
5-R05 Individual devices for applicable Jury members plus President controls.	Compliant	D.1.5	Jury controls for jury members are the same as the referee controls, but without the warning features. Referee controllers can be used as backups for jury controllers (the warnings would simply not be emitted). President controls are of two kinds. The solution includes a physical keypad with all the required real-time referee decision display and the required jury member behaviour (green until all decisions visible) A complementary tablet display is available in the kit; this enables the jury president to enter the reason for jury reversals.
5-R06 Authoritative timer/controller with	Compliant	D.1.6	Timing is done on the main server using the operating system timers. Operating system timing resolution is 1 ms as used. Propagation times of MQTT messages is under 10

Requirement	Compliance	Reference	Justification
approved displays and audio.			ms. Time is displayed on the attempt board and on an athlete-facing clock.
5-R07 Competition and warm-up presentation with mandatory athlete/attempt/weight /clock/decision data.	Compliant	D.1.7	The attempt board is any standard FullHD TV with HDMI inputs. The kit does not include the actual TV screens, that are assumed to be provided locally. Two computers provide attempt board output. One provides a standard black background output that is replicated on all TVs in the waiting, marshal and warmup areas. The other output is used for the attempt boards in the main room, where some styling (e.g., changing the background color) may have been done. The kit includes splitters and extenders, assuming an attempt board is present in each of the waiting, marshal, and warm-up rooms, and that these areas are immediately behind the competition stage. Further splitting or extension of the attempt board HDMI signal to reach multiple attempt boards is to be done using equipment and cabling provided locally. For the main room, it is assumed that two attempt boards are present, one on each side of the stage.
5-R08 Real-time competition and warm-up displays with correct ranking and state.	Compliant	D.1.8	The situation for the scoreboards is the same as for attempt boards. There are two HDMI sources, one for the stage area, one for the back. For the stage area, the signal is assumed to go to a single large LED scoreboard.
5-R09 Approved cameras, recording/replay and Jury output for events requiring VPT.	Compliant	D.1.9	The VPT system is under the full control of the competition staff, and the TCRR-mandated angles can be displayed in self-service mode by the jury if so required (or provided by the competition director). As indicated in TCRR, the replay clips are accessible by the venue video team and broadcast teams. Such access will be controlled by networking equipment configurations. The VPT system relies on four PTZ cameras. Based on our experience in Continental competitions, we propose Tenveo PTZ 60fps cameras. These cameras provide Power-over-Ethernet RTSP video feeds, HDMI, and compressed H.264 USB feeds. OWLCMS has a VPT module that automatically clips and attributes the replays to the athlete lifting. This module keeps a full indexed archive of all the lifts, at all angles. The VPT will execute on a high-performance laptop that also meets the requirements for the vMix software. If a client federation wants to add such additional video-processing capabilities, the proposed equipment will be sufficient for them to use their own license on the video-processing laptop.
5-R10 Documented, segmented, monitored event network with common time.	Compliant	D.1.10	The networking configuration is based on switching. Unmanaged switches are sufficient. A minimum of 4 POE (Power-over-Ethernet) ports is necessary to provide power to the networked PTZ cameras. A 16-port switch per platform is provided. An ASUS WiFi router is included. The router is configured to

Requirement	Compliance	Reference	Justification
			provide DHCP addresses to all machines in the kit. The only machines whose addresses matter are the main OWLCMS server and the VPT laptop on each platform. These will be set statically. WiFi is not normally used for the competitions, only as a convenience during setup. If federation personnel request access, it is preferred to loan them a network dongle. This eliminates any concern about illicit WiFi access. In the proposed configuration, the number of networking switches is kept to a minimum to prevent illicit connections. All the machines on one platform are connected to the switch on that platform. Use of a switch also ensures that video traffic from the cameras goes directly to the video equipment and does not touch other machines that don't need it OWLCMS networking traffic is minimal. The application is routinely used in a cloud configuration with very remote servers and no perceptible delay.
5-R11 Controlled read-only path from accepted state to approved consumers.	Compliant	D.1.11	The video laptop will host all the external interfaces. In actual practice, sending information out to a games central is very low usage (and ODF protocols are push only). OWLCMS can also publish to cloud-based scoreboards and other feature, but those are also push only. In the past, outside broadcasters have used the OVR-provided screen layouts directly. This was accommodated by opening a port on the router to a specifically white-listed address. This capability is generic with the equipment provided.
5-R12 Reliable official documents with fallback.	Compliant	D.1.12	A Windows laptop with Microsoft Office is used for data entry and report generation to maximize compatibility with whatever printer is made available. OWLCMS produces documents in Excel format. Documents should be exported as PDF in non-editable mode and kept as such for archival. Start books, Result books, Refereeing schedules and similar sophisticated layouts are produced as PDF.
5-R13 Protection and approved autonomy for critical equipment.	Compliant	D.1.13	A UPS will be provided to protect the main server and the networking switches. The rest of the equipment, such as TVs, will likely go down during power issues, so there is no way to prevent outages other than a full generator (as used in Continental championships held in tropical countries subject to power fluctuations).
5-R14 Health, service, device, storage, time and stale-state monitoring.	Compliant	D.1.14	The OWLCMS software has very low memory and disk I/O requirements. Use of the normal operating system indicators is therefore sufficient. An administrative function does exist to query more precise runtime information and add it to the technical log, but only ever used during stress tests to confirm correct operation.
5-R15 Rapidly deployable replacements for critical equipment families.	Compliant	D.1.15	All the stations only require the presence of a current browser (Chrome, Edge, Firefox or Safari), and for some of them, the ability to connect a USB device. So essentially any laptop can be used, whether Windows, macOS, Linux.

Requirement	Compliance	Reference	Justification
			Chromebooks can be used when no USB device is needed. OWLCMS runs on Windows, macOS,
5-R16 Safe, labelled cabling and equipment placement without obstruction or sightline conflicts.	Compliant	D.1.16	[To be completed]
5.1 For each licensed event, the supplier shall maintain make/model, quantity, serial or asset ID, firmware, location, connections, power source, owner, health status, spare path and approved deviations.	Compliant	D.1.17	For each licensed event, the supplier shall maintain make/model, quantity, serial or asset ID, firmware, location, connections, power source, owner, health status, spare path and approved deviations. Supplier response [TO BE COMPLETED]

RFP Section 6

Requirement	Compliance	Reference	Justification
6.1-R01 Implement the effective IWF rule pack and support controlled updates.	Compliant	C.1.1.1	The solution is tested against the TCRR rules using automated tests. Any new release is tested against regressions.
6.1-R02 Competition-critical functions shall not depend on unavailable internet, licence-server or payment services.	Compliant	C.1.1.2	Once installed, the solution is completely autonomous and works with no internet connection whatsoever
6.1-R03 Exact production version, configuration, database schema, rule pack and relevant assets shall be identifiable and reproducible.	Compliant	C.1.1.3	Releases are created by a fully repeatable automated build process, and under strict version control. This applies both to OWLCMS and the Blue Owl components. Any past version can be retrieved and matched against its source code.
6.1-R04 Installation and rollback shall be documented and repeatable.	Compliant	C.1.1.4	Installation and rollback are done with an installation program that allows rollbacks. Within a major version, rollback to a previous version is always possible. Rollback to a prior major version is possible, obviously losing features introduced in that version.
6.1-R05 Automatic updates shall not occur during an event freeze unless approved.	Compliant	C.1.1.5	The system does not perform any type of automatic update. Operating system updates shall be managed by IWF or event personnel and not part of the product.
6.2-R01 Every sporting fact shall have one declared authoritative owner.	Compliant	C.1.2.1	Any change to the competition state is uniquely attributed to a human actor, to a device input, or to an automated calculation. The audit log reflects this attribution.
6.2-R02 Derived screens, feeds, documents and caches shall be traceable to the accepted state.	Compliant	C.1.2.2	All computed results are directly attributable to programmed rules based on the authoritative data.
6.2-R03 Corrections shall be authorised, atomic, attributable and propagated.	Compliant	C.1.2.3	Once the competition has started, all changes are meant to be made using the interactive interfaces or the explicit import features of the programs such that audit traces are kept. Database state that does not correspond to audited actions is evidence of tampering. Should intentional exceptional corrective actions by the supplier requiring database queries or similar be necessary, the corresponding authorizations are expected to be managed separately as part of incident response.
6.2-R04 Duplicate, lost, conflicting or out-of-order transactions	Compliant	C.1.2.4	The system applies strict serialization of transactions. A relational database is used to enforce integrity constraints.

Requirement	Compliance	Reference	Justification
shall be prevented or detected.			
6.2-R05 A complete machine-readable event export shall be available to IWF.	Compliant	C.1.2.5	The database export contains all the event data. The audit logs are also machine-readable.
6.3 Material interfaces shall document producer/consumer, purpose, version, schema/field mapping, identifiers, status/null rules, sequencing/revision, time source, latency, authentication, acknowledgement, retry/reconnect, monitoring and fallback.	Compliant	G.1	Material interfaces shall document producer/consumer, purpose, version, schema/field mapping, identifiers, status/null rules, sequencing/revision, time source, latency, authentication, acknowledgement, retry/reconnect, monitoring and fallback.
6.4 All third-party software, libraries, services, licences and hardware necessary for operation shall be declared, including version, ownership, support status and cloud/public-internet dependency.	Compliant	C.2	All third-party software, libraries, services, licences and hardware necessary for operation shall be declared, including version, ownership, support status and cloud/public-internet dependency. All dependencies used in OWLCMS and Blue Owl are public and available on public servers.

RFP Section 7

Requirement	Compliance	Reference	Justification
7.1-R01 Least-privilege access by role and device identity.	Compliant	F.1.1.1	OWLCMS supports separate logins for every role, and where relevant, separate logins for each role and platform. Devices are identified per platform and authenticated using the MQTT identity of the MQTT interface.
7.1-R02 Separation of competition, device, public/display, database and management functions where applicable.	Compliant	F.1.1.2	The role structure separates the technical official roles, preparation roles, the competition director, and admin functions.
7.1-R03 Protection of credentials, secrets and data in transit/at rest where required.	Compliant	F.1.1.3	All credentials or secrets are stored encrypted and transported in encrypted form. They are never written to log files in clear. Account passwords use PBKDF2 encryption with HMAC-SHA256. Todo: tracker connection in database needs propping up.
7.1-R04 Controlled remote support, removable media and patch/deployment processes.	Compliant	F.1.1.4	Security options are in the database or in external configuration files; updating to a fixed version is independent of the options. All future versions of OWLCMS will support the security requirements of IWF.
7.1-R05 Tamper-resistant audit sufficient to reconstruct material actions.	Compliant	F.1.1.5	The audit log is tamper-resistant and any tampering is detectable (the logger keeps its own copy of the lines and cryptographically signs blocks of lines). The audit logs includes every single modification to the athlete and lifting data, and all events coming from devices (decisions etc.)
7.1-R06 No public or anonymous path may write authoritative competition state.	Compliant	F.1.1.6	The only writing to the database is done by authenticated users. The database is the only source of written archival, that are also cryptographically signed.
7.2 Core competition, officiating, timing, local displays and local official documents shall remain usable during loss of public internet or cloud connectivity.	Compliant	F.1.2	Core competition, officiating, timing, local displays and local official documents shall remain usable during loss of public internet or cloud connectivity. The normal mode of operation of OWLCMS is completely off-line, on a private local area network, with all key components wired (in particular the refereeing functions).
7.3-R01 Protected backup copies of data, configuration, rules, assets and audit.	Compliant	F.1.3.1	All files that constitute the OWLCMS state are in a single folder (with the exception of cryptographic key) Any means of backup can be used – normally copying the folder to a removable device or off-site storage.
7.3-R02 Timed restoration demonstrating the approved recovery objective.	Compliant	F.1.3.2	Restoration of a saved state from scratch normally takes less than a minute.
7.3-R03 Defined promotion, split-brain	Compliant	F.1.3.3	The system would not operate in tandem. Checkpoint copies would be used.

Requirement	Compliance	Reference	Justification
prevention, failback and reconciliation where redundancy is claimed.			
7.3-R04 Cold-start and rollback procedures using controlled recovery media.	Compliant	E.1.3.4	Checkpoint copies of the export format can be obtained on demand, and constitute full rollback points. Restarting the system without a database file creates a minimal empty one that can be immediately loaded from a checkpoint.
7.4 A manual Protocol and rehearsed manual-continuity procedure shall remain available. Staff shall know when to continue, when to STOP and how to reconcile manual records with the electronic state.	Compliant	E.1.4	A manual Protocol and rehearsed manual-continuity procedure shall remain available. Staff shall know when to continue, when to STOP and how to reconcile manual records with the electronic state. This is standard operational procedure. A power outage, or a networking mishap can prevent use the system and cause a revert to a “manual cards and flags” mode. OWLCMS has been tested in such situations before. As soon as the system is restored, someone can simply copy the information as it stands on the cards, even while lifting is going on that platform. As soon as the information has been updated and matches the paper trail, the screens can be turned on for that platform. All the lifting order and such is derived from the authoritative card data as entered. Writing operational procedures as expected by IWF would take place before deployment.
7.5 Actionable monitoring shall cover critical services, devices, timing, storage, network and stale-state conditions, with clear escalation responsibilities.	Compliant	E.1.5	Actionable monitoring shall cover critical services, devices, timing, storage, network and stale-state conditions, with clear escalation responsibilities. The actual OTIS kit requires a router and, for each platform, a switch. As part of the certified kit, this constitutes the monitoring boundary. The kit devices are attached directly to these and can be monitored (including remote logins).

RFP Section 8

Requirement	Compliance	Reference	Justification
8.1-R01 Legally established entity able to contract with IWF.	Compliant	A.1.1.1	[To be completed]
8.1-R02 Relevant experience delivering competition scoring/results or comparable mission-critical systems.	Compliant	A.1.1.2	[To be completed]
8.1-R03 Ability to accept end-to-end responsibility for the complete system and declared third-party components.	Compliant	A.1.1.3	[To be completed]
8.1-R04 Qualified technical, competition-operations and support personnel.	Compliant	A.1.1.4	[To be completed]
8.1-R05 International training, event support and incident-response capability.	Compliant	A.1.1.5	[To be completed]
8.1-R06 Ability to maintain the approved baseline for the license period.	Compliant	A.1.1.6	[To be completed]
8.2-R01 Provide accurate product and hardware baseline.	Compliant	H.1.1	[To be completed]
8.2-R02 Provide complete technical submission and evidence package.	Compliant	H.1.2	[To be completed]
8.2-R03 Make the declared system available for IWF testing and demonstrations.	Compliant	H.1.3	[To be completed]
8.2-R04 Correct deficiencies and complete regression testing when required.	Compliant	H.1.4	[To be completed]
8.2-R05 Notify IWF of material product, security, hardware,	Compliant	H.1.5	[To be completed]

Requirement	Compliance	Reference	Justification
interface or rule-logic changes.			
8.2-R06 Maintain qualified support personnel and appropriate spares.	Compliant	H.1.6	[To be completed]
8.2-R07 Protect IWF, athlete and event information and comply with applicable data-protection requirements.	Compliant	H.1.7	[To be completed]

RFP Section 9

Requirement	Compliance	Reference	Justification
9.1-R01 Acceptance of the IWF TCRR as the controlling sporting rules.	Compliant	K.3.1	[TO BE COMPLETED]
9.1-R02 Identification of all material limitations and dependencies.	Compliant	K.3.2	[TO BE COMPLETED]
9.1-R03 Ability to reproduce the tested system from controlled installation material.	Compliant	K.3.3	[TO BE COMPLETED]
9.1-R04 Offline operation and tested recovery for the claimed profile.	Compliant	K.3.4	[TO BE COMPLETED]
9.1-R05 No automatic licensing mechanism that can interrupt an active competition.	Compliant	K.3.5	[TO BE COMPLETED]

RFP Section 10

Requirement	Compliance	Reference	Justification
10-S01 Completeness, eligibility and declarations.	Compliant	I.1.1	[To be completed]
10-S02 Architecture, functionality, hardware, interfaces, security and resilience.	Compliant	I.1.2	[To be completed]
10-S03 Freeze baseline, profile, capacity, dependencies and test plan.	Compliant	I.1.3	[To be completed]
10-S04 Controlled normal, edge, security, capacity and failure testing.	Compliant	I.1.4	[To be completed]
10-S05 Resolve deficiencies and provide regression evidence.	Compliant	I.1.5	[To be completed]
10-S06 Representative event tempo with operators, officials, displays, feeds and recovery.	Compliant	I.1.6	[To be completed]
10-S07 Review requirements, hard gates, evidence and findings.	Compliant	I.1.7	[To be completed]
10-S08 Complete legal and operational conditions.	Compliant	I.1.8	[To be completed]
10-S09 Check actual installation against approved baseline.	Compliant	I.1.9	[To be completed]
10-S10 Review incidents, changes, support and expiry.	Compliant	I.1.10	[To be completed]
10.1 Evidence shall be attributable to a specific baseline and test. Screenshots alone are insufficient where logs, physical inspection or live behaviour are necessary.	Compliant	I.1.11	Evidence shall be attributable to a specific baseline and test. Screenshots alone are insufficient where logs, physical inspection or live behaviour are necessary. Supplier response [TO BE COMPLETED]

RFP Section 11

Requirement	Compliance	Reference	Justification
HG-01 Incorrect sporting fact, rank, medal, record or rule-critical transition.	Compliant	1.2.1	[To be completed]
HG-02 Split, duplicated, lost or silently overwritten accepted state.	Compliant	1.2.2	[To be completed]
HG-03 Unsafe or ambiguous timing, decision, Down signal or restart.	Compliant	1.2.3	[To be completed]
HG-04 Unauthorised, partial, untraceable or unreconciled correction.	Compliant	1.2.4	[To be completed]
HG-05 Uncontrolled loss of competition or unverified recovery.	Compliant	1.2.5	[To be completed]
HG-06 Uncontrolled privilege, public write path or material compromise.	Compliant	1.2.6	[To be completed]
HG-07 Contradictory outputs or untraceable final publication.	Compliant	1.2.7	[To be completed]
HG-08 Unknown tested version, missing evidence or irreproducible installation.	Compliant	1.2.8	[To be completed]
HG-09 Missing critical equipment, unsafe installation or unresolved obstruction.	Compliant	1.2.9	[To be completed]
11.1-R01 All applicable mandatory requirements and tests must be disposed of.	Compliant	1.2.10.1	[To be completed]
11.1-R02 Applicable TCRR requirements and hard gates must pass.	Compliant	1.2.10.2	[To be completed]
11.1-R03 No open Critical finding may remain.	Compliant	1.2.10.3	[To be completed]

Requirement	Compliance	Reference	Justification
11.1-R04 Traceability from requirement to evidence and test must be complete.	Compliant	I.2.10.4	[To be completed]
11.1-R05 Successful FAT, recovery and operational validation are required for the claimed profile.	Compliant	I.2.10.5	[To be completed]
11.1-R06 Partial is not Pass.	Compliant	I.2.10.6	[To be completed]
11.1-R07 A profile limitation cannot change an IWF sport rule or bypass a hard gate.	Compliant	I.2.10.7	[To be completed]

RFP Section 12

Requirement	Compliance	Reference	Justification
12.1 The IWF licence shall be valid for a four-year period. The certificate shall state supplier, product/baseline, profiles, capacities, limitations, deviations, validity and surveillance conditions.	Compliant	K.1.1	The IWF licence shall be valid for a four-year period. The final contract shall state the exact licence duration and renewal conditions. Supplier response [TO BE COMPLETED]
12.2-R01 Controlled update and review.	Compliant	K.1.2.1	[TO BE COMPLETED]
12.2-R02 Supplier assessment and IWF notification where applicable.	Compliant	K.1.2.2	[TO BE COMPLETED]
12.2-R03 Impact assessment and approved regression/delta test.	Compliant	K.1.2.3	[TO BE COMPLETED]
12.2-R04 Re-homologation before event use.	Compliant	K.1.2.4	[TO BE COMPLETED]
12.2-R05 Named approval, backup, rollback, targeted test and retrospective evidence.	Compliant	K.1.2.5	[TO BE COMPLETED]
12.3-R01 Only the approved baseline and event configuration may be used.	Compliant	H.2.1	[To be completed]
12.3-R02 Actual hardware, software, network, power, displays, spares and staffing must match the approved event design.	Compliant	H.2.2	[To be completed]
12.3-R03 Dress rehearsal and failure/recovery test shall be completed before readiness sign-off.	Compliant	H.2.3	[To be completed]
12.3-R04 Substitutions require approved	Compliant	H.2.4	[To be completed]

Requirement	Compliance	Reference	Justification
deviation or delta testing before use.			
12.4-R01 Defined escalation contacts and event support.	Compliant	H.3.1	[To be completed]
12.4-R02 Critical incidents documented, contained, investigated and reported.	Compliant	H.3.2	[To be completed]
12.4-R03 Serious incidents, unauthorised changes, false evidence, unsupported baselines or material security compromise may lead to restriction, suspension or withdrawal.	Compliant	H.3.3	[To be completed]
12.5 IWF shall receive the rights, documentation, exports, interfaces and operational access necessary to use, verify, archive and maintain the approved system and competition records.	Compliant	K.2	The supplier retains its intellectual property except where otherwise agreed. IWF shall receive the rights, documentation, exports, interfaces and operational access necessary to use, verify, archive and maintain the approved system and competition records. Final contractual wording will be agreed with the selected supplier. Supplier response [TO BE COMPLETED]

RFP Section 13

Requirement	Compliance	Reference	Justification
13-R01 The Licensing Fee for the four-year period is US\$20,000, payable on an annual basis of US\$5,000 per year.	Compliant	J.1.1	US\$5,000 per year / US\$20,000 total Assumptions: [TO BE COMPLETED]
13-R02 Licence	Compliant	J.1.2	[TO BE COMPLETED] Assumptions: [TO BE COMPLETED]
13-R03 Implementation	Compliant	J.1.3	[TO BE COMPLETED] Assumptions: [TO BE COMPLETED]
13-R04 Equipment	Compliant	J.1.4	[TO BE COMPLETED] Assumptions: [TO BE COMPLETED]
13-R05 Support / maintenance	Compliant	J.1.5	[TO BE COMPLETED] Assumptions: [TO BE COMPLETED]
13-R06 Optional services	Compliant	J.1.6	[TO BE COMPLETED] Assumptions: [TO BE COMPLETED]
13-R07 Recurring charges	Compliant	J.1.7	[TO BE COMPLETED] Assumptions: [TO BE COMPLETED]

Appendices

Appendix A. IWF responsibilities

The supplier acknowledges the IWF responsibilities stated in section 8.3 of the RFP.

Comments, if any

[TO BE COMPLETED]

DRAFTING NOTE - DELETE BEFORE SUBMISSION

Before PDF export, replace all bracketed text, remove all drafting notes, and check that the proposal remains one coherent package.